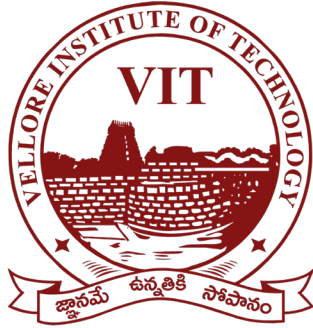




VIT-AP UNIVERSITY

NASA KOI EXOPLANET CLASSIFICATION USING DIFFERENT ML ALGORITHMS

Course:- CSE2019-Machine Learning Theory

Professor:- Vasavi Sanikommu

Slot:- A1+TA1

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Git Hub Repository Link:-

<https://github.com/Likhith2062/ExoPlanetDetection>

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NASA KOI EXOPLANET CLASSIFICATION USING MACHINE LEARNING

1. Team Details & GitHub Repository

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2. Introduction

Exoplanet detection is an important classification problem because observations collected by astronomical surveys contain many signals that resemble planetary transits without necessarily representing planets. The project investigates whether numerical measurements associated with Kepler Objects of Interest can be used to distinguish confirmed exoplanets from false-positive candidates. Rather than treating the task as a purely theoretical machine-learning exercise, the work follows a practical experimental pipeline: obtain the NASA dataset, preprocess the numerical attributes, compare several supervised learning algorithms, select a strong baseline, and then optimize the selected model while keeping the evaluation procedure consistent.

2.1 Problem Statement

The central problem is to construct a binary machine-learning classifier that determines whether a Kepler Object of Interest should be classified as an exoplanet or as a non-exoplanet or false-positive case. The available observations contain numerical measurements describing properties of the transit signal and the host star. The challenge is therefore to learn a reliable decision boundary from these attributes while avoiding an evaluation procedure that makes the model appear better simply because it was tested on observations it had already seen during training. The project uses stratified cross-validation to address this evaluation requirement.

2.2 Motivation

The motivation for the project comes from the practical consequence of classification errors in astronomical candidate screening. A false negative can cause a potentially interesting planetary candidate to be overlooked, while excessive false positives can consume attention and follow-up resources. Consequently, accuracy alone is not sufficient to describe the usefulness of the classifier. Recall is particularly important because it measures the fraction of actual exoplanets that the system successfully identifies. Precision and F1-score are also considered so that improving recall does not come at an unreasonable cost in prediction quality. This motivates later threshold optimization.

3. Dataset Description

The project uses data obtained from the NASA Exoplanet Archive, specifically the cumulative Kepler Objects of Interest data source supplied for the experiments. The working dataset contains numerical measurements associated with KOI observations and a categorical disposition used as the prediction target. During preprocessing, the selected numerical attributes and target were retained, missing records were removed, and the target was represented as a binary classification suitable for the machine-learning algorithms. The final processed dataset used by the experiments contains 6,625 usable observations, comprising 3,895 non-exoplanet cases and 2,730 exoplanet cases.

4. Problem Analysis & Proposed Approach

The project was designed as an experimental classification pipeline rather than as a single one-shot model-training script. The workflow begins with the raw or previously processed dataset and applies a consistent preprocessing stage before training. Multiple supervised algorithms are then evaluated under the same ten-fold stratified cross-validation procedure. The strongest candidate is selected using a combination of accuracy, precision, recall, F1-score, and ROC-AUC rather than one metric in isolation. Random Forest was subsequently optimized through controlled experiments on classification threshold, number of estimators, and maximum tree depth. This approach makes each design decision traceable to measured results.

5. Data Preprocessing

The preprocessing stage converts the downloaded astronomical table into a compact machine-learning dataset containing only the variables required by the project. Records with missing values in the selected attributes or target are removed so that the downstream algorithms receive complete numerical observations. The target disposition is converted into a binary representation in which confirmed exoplanets form the positive class and false-positive observations form the negative class. Keeping preprocessing explicit and repeatable is important because all model comparisons must use the same observations and feature definitions. The resulting processed CSV becomes the stable input used throughout model selection and Random Forest optimization.

6. Feature Selection

Feature selection was intentionally kept compact because the project focuses on understanding the effect of a manageable set of physically meaningful numerical measurements rather than introducing a large automated feature-engineering pipeline. The final classifier uses nine attributes: `koi_depth`, `koi_duration`, `koi_impact`, `koi_model_snr`, `koi_max_mult_ev`, `koi_srad`, `koi_smass`, `koi_steff`, and `koi_bin_oedp_sig`. These variables represent transit-signal measurements and stellar properties available in the KOI dataset. Using the same nine variables for every candidate algorithm ensures that differences in performance primarily reflect the learning methods rather than unequal access to input information.

6.1 Selected Features

The selected feature set consists of `koi_depth`, `koi_duration`, `koi_impact`, `koi_model_snr`, `koi_max_mult_ev`, `koi_srad`, `koi_smass`, `koi_steff`, and

koi_bin_oedp_sig. Together these measurements provide information about transit depth, transit duration, impact geometry, signal strength, multiple-event significance, and stellar characteristics. The target variable is koi_disposition after conversion to the binary labels required by the classifier. The feature list is kept explicitly in the main training workflow so that the same columns are used for preprocessing, cross-validation, final training, and later visualization, reducing the possibility of accidental feature mismatch.

- koi_depth
- koi_duration
- koi_impact
- koi_model_snr
- koi_max_mult_ev
- koi_srad
- koi_smass
- koi_steff
- koi_bin_oedp_sig

6.2 Reasoning Behind Selection

The selected variables were retained because they directly describe measurable properties of the observed transit and its associated star, making them suitable numerical inputs for supervised classification. The project deliberately avoids adding a large number of unrelated attributes simply to increase model complexity. This keeps the experimental interpretation clearer: every algorithm receives the same nine numerical features and the same target definition. The approach also makes the final Random Forest easier to reproduce and explain. Feature selection was therefore guided by availability, numerical suitability, relevance to the KOI classification task, and consistency across all experiments.

Feature Name

koi_depth

koi_duration

koi_impact

koi_model_snr

koi_max_mult_ev

Reason for Selection

Measures the depth of the observed transit and therefore captures how strongly the transit signal affects the measured stellar brightness.

Represents the duration of the transit, providing information about the timing and geometry of the observed transit event.

Describes the transit impact parameter and therefore provides information about the geometry of the object passing in front of the star.

Provides the signal-to-noise strength of the modeled transit, helping distinguish stronger and more reliable transit signals from weak observations.

Captures the maximum multiple-event

	significance, providing evidence about the consistency and significance of repeated transit events.
koi_srad	Provides the stellar radius, adding host-star physical context that can help interpret the observed transit characteristics.
koi_smass	Provides the stellar mass, supplying additional physical information about the host star associated with the observed candidate.
koi_steff	Represents stellar effective temperature and contributes information about the physical characteristics of the host star.
koi_bin_oedp_sig	Provides a significance measure associated with the odd/even transit-depth comparison, which is useful when distinguishing planetary transit patterns from other signals.

7. Experimental Methodology

All candidate models were evaluated using ten-fold stratified cross-validation with shuffling and a fixed random state of 42. Stratification preserves the relative representation of the two target classes across folds, which is important for a binary dataset where the classes are not perfectly balanced. The same evaluation structure was retained when Random Forest was optimized. This means that changes in threshold or hyperparameters were judged using comparable experiments instead of changing the validation procedure between tests. The final model was trained on the complete processed dataset only after the selected configuration had been established through the validation experiments.

7.1 10-fold Stratified Cross Validation

In ten-fold stratified cross-validation, the dataset is divided into ten folds while maintaining class proportions as closely as possible. Nine folds are used to train a temporary model and the remaining fold is used for evaluation, with the process repeated until every observation has served as validation data once. The reported metric is the mean of the ten fold scores, accompanied by the standard deviation. This provides a more stable estimate than relying on a single train-test split and also produces out-of-fold predictions that can be used for final ROC and confusion-matrix visualization without evaluating a model on its own training observations.

7.2 Evaluation Metrics

Five metrics are used throughout the project. Accuracy measures the overall proportion of correct predictions. Precision measures how many observations predicted as exoplanets are actually exoplanets. Recall measures how many actual exoplanets are successfully detected and is particularly important because missed candidates are undesirable in the intended application. F1-score is the harmonic mean

of precision and recall and therefore summarizes their balance. ROC-AUC measures the classifier's ability to rank positive observations above negative observations across decision thresholds. Considering all five prevents the model-selection decision from depending on a single number.

7.3 Reproducibility

Reproducibility is supported through fixed random seeds, an explicitly defined feature list, a fixed ten-fold stratified cross-validation configuration, and recorded model hyperparameters in the performance JSON files. The experiments also preserve the individual fold scores instead of recording only the final average, allowing variation between folds to be inspected later. The Random Forest configuration ultimately used 165 estimators, a maximum depth of 30, the Gini criterion, square-root feature selection, and `random_state` 42, with the classification threshold set independently to 0.46. These settings form the final reproducible configuration reported by the project.

8. Model Selection

The initial model-selection stage compares nine supervised learning algorithms under the same data and validation conditions: AdaBoost, CART Decision Tree, Extra Trees, Gaussian Naive Bayes, Gradient Boosting, K-Nearest Neighbours, Logistic Regression, Random Forest, and Support Vector Machine. The supplied comparison graphs show that Random Forest achieves the strongest overall combination in the initial tests, including the highest accuracy at 90.82%, the highest F1-score at 88.99%, and the highest ROC-AUC at 96.39%. Extra Trees and Gradient Boosting are competitive, while Naive Bayes produces very high recall but considerably weaker precision. These results justify selecting Random Forest for further optimization.

8.1 Accuracy Graph

The accuracy comparison shows Random Forest at 90.82%, higher than the other tested algorithms in the initial model-selection experiment. Extra Trees reaches 90.28% and Gradient Boosting reaches 90.02%, demonstrating that several tree-based methods perform well on the selected numerical attributes. The difference is not enormous, which is useful evidence that the dataset contains learnable structure across several algorithms. Nevertheless, Random Forest provides the highest measured accuracy while also performing strongly on the other evaluation measures. The graph therefore contributes to the decision without being treated as the only criterion for model selection.

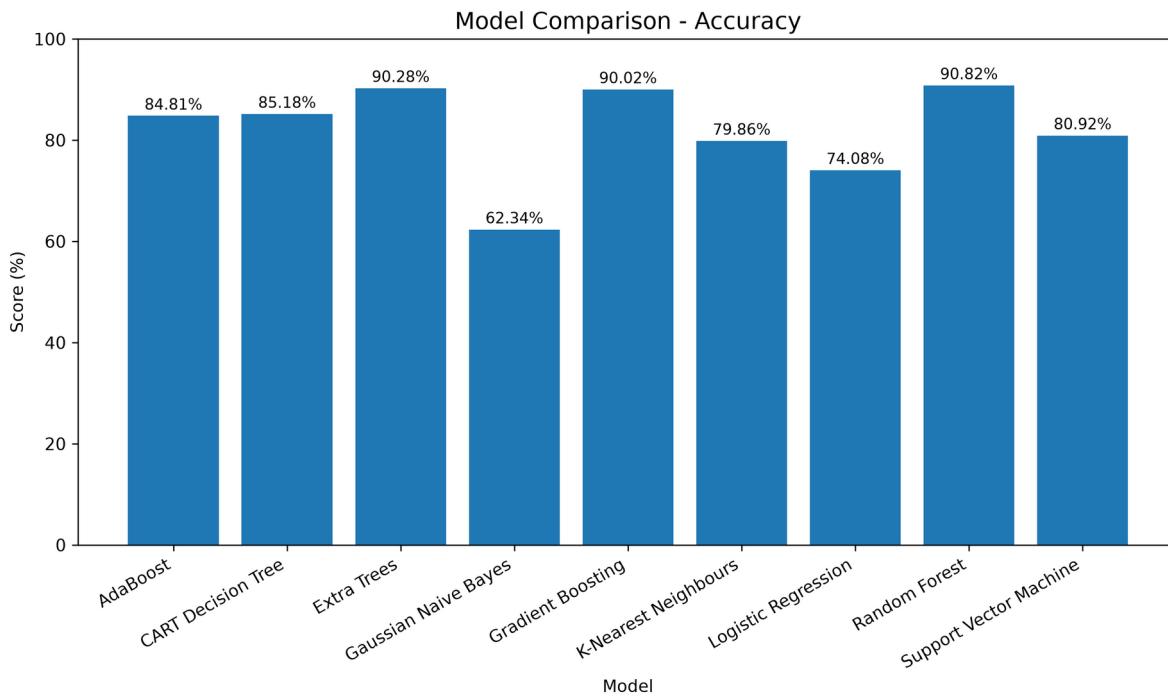


Figure 1. Initial model-selection comparison: Accuracy Graph.

8.2 Precision Graph

The precision comparison places Random Forest at 88.03%, the highest value among the nine algorithms in the initial test. Extra Trees follows at 87.15%, while Gradient Boosting reaches 85.27%. Precision is important in this application because a model that labels too many non-exoplanet observations as exoplanets would generate unnecessary candidate follow-up. Random Forest therefore provides a strong starting point from the perspective of controlling false-positive classifications. At the same time, the project does not maximize precision blindly, because the later experiments deliberately accept a modest precision reduction when it produces a meaningful improvement in recall.

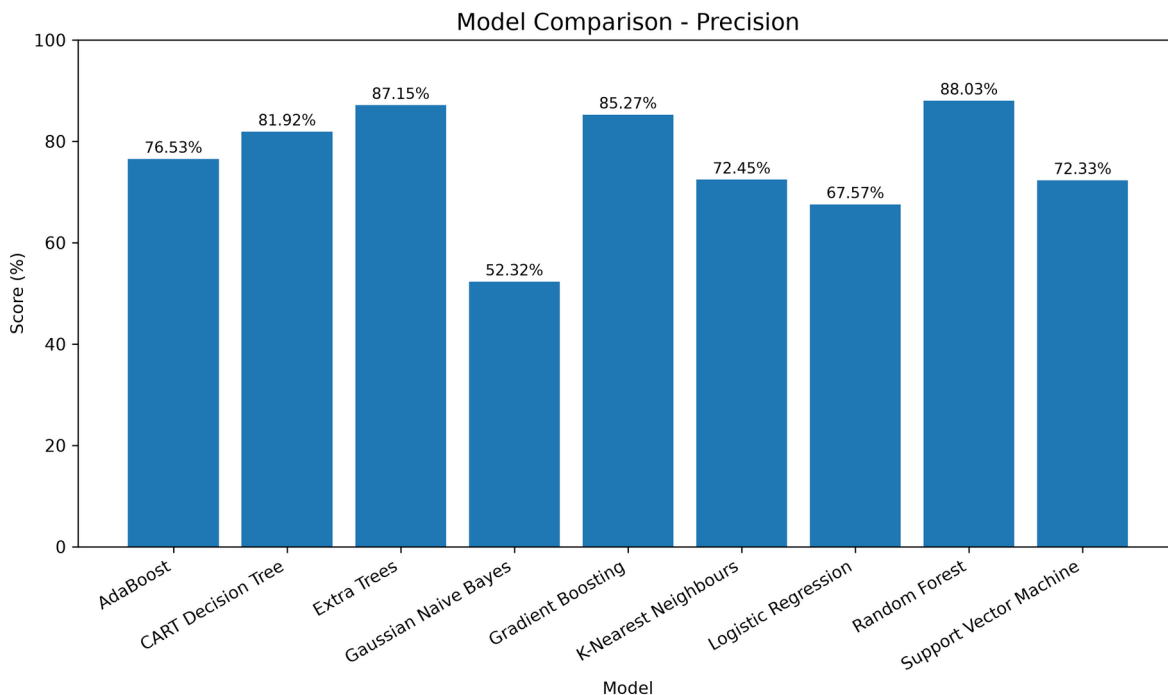


Figure 2. Initial model-selection comparison: Precision Graph.

8.3 Recall Graph

The recall comparison highlights an important trade-off. Gaussian Naive Bayes achieves the highest recall at 97.40%, while Random Forest reaches 90.00% in the initial configuration. AdaBoost and Gradient Boosting are also close to the Random Forest result. Although Naive Bayes detects a larger fraction of positive cases, its precision is much lower, meaning that the additional positives come with substantially more false alarms. Random Forest therefore represents a more balanced candidate. The later threshold experiment specifically investigates whether its decision boundary can be shifted to recover more true exoplanets without sacrificing the overall balance excessively.

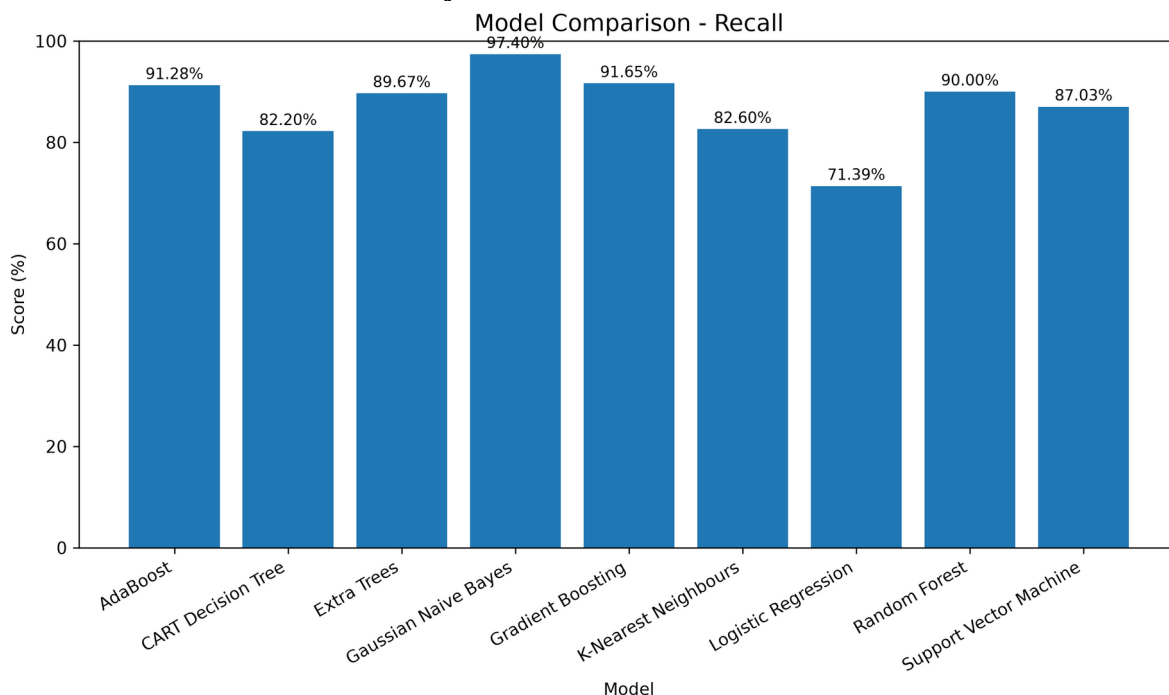


Figure 3. Initial model-selection comparison: Recall Graph.

8.4 F1 Graph

The F1-score comparison is particularly useful because it combines precision and recall into a single measure of balance. Random Forest achieves the highest initial F1-score at 88.99%, slightly ahead of Extra Trees at 88.38% and Gradient Boosting at 88.33%. This result supports the decision to continue with Random Forest because it does not simply win on accuracy while performing poorly on another important metric. Instead, it provides the strongest compromise between identifying exoplanets and avoiding unnecessary positive classifications. F1-score later becomes a central criterion when selecting the final classification threshold.

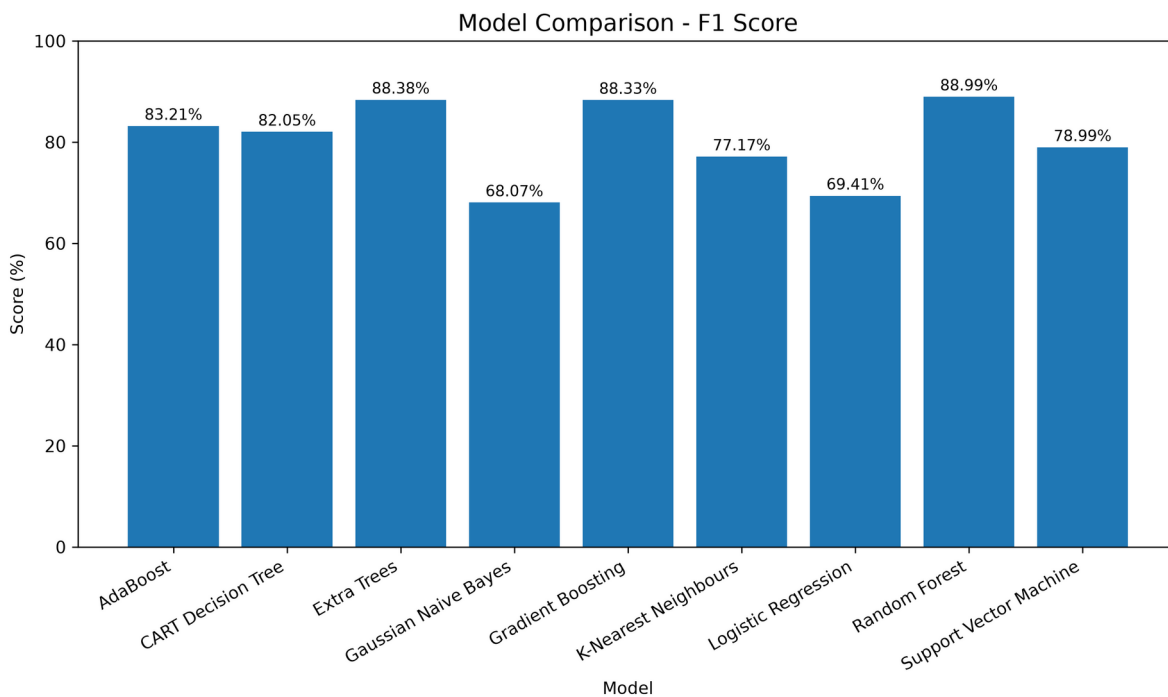


Figure 4. Initial model-selection comparison: F1 Graph.

8.5 ROC-AUC Graph

The ROC-AUC comparison shows that Random Forest reaches 96.39%, the highest value among the nine tested algorithms. Extra Trees and Gradient Boosting are close at 96.17% and 96.19%, respectively, while several other models are noticeably lower. ROC-AUC is useful here because it evaluates ranking ability across many possible thresholds rather than committing to the default 0.50 decision boundary. The high Random Forest AUC indicates that the model generally separates the two classes well, giving the project room to adjust the operating threshold later for the application's recall requirements.

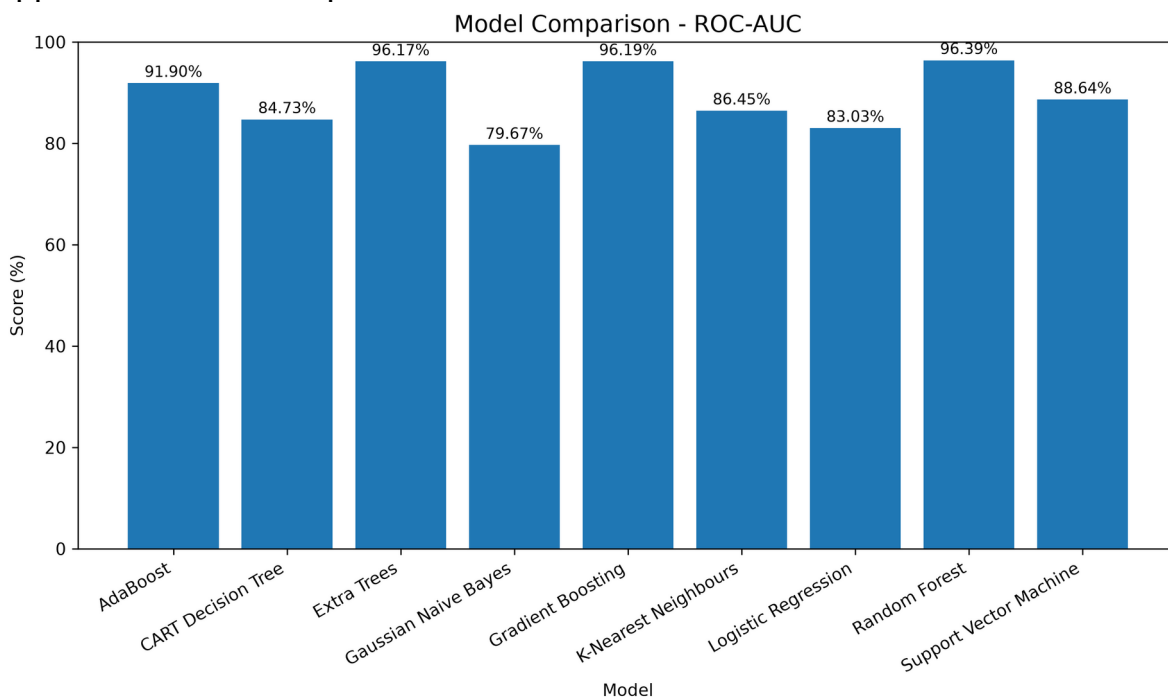


Figure 5. Initial model-selection comparison: ROC-AUC Graph.

8.6 Why Random Forest Was Selected

Random Forest was selected because its advantage was not restricted to one isolated metric. It produced the highest initial accuracy, precision, F1-score, and ROC-AUC in the supplied model-comparison results, while maintaining a respectable recall of 90.00%. Its ensemble of decision trees also makes it a natural candidate for tuning through parameters such as the number of estimators and maximum tree depth. This combination of measured performance and practical tunability made Random Forest the most suitable model to carry into the optimization phase. More complex ensemble combinations were not necessary because the strongest individual model was already highly competitive.

9. Random Forest Optimization

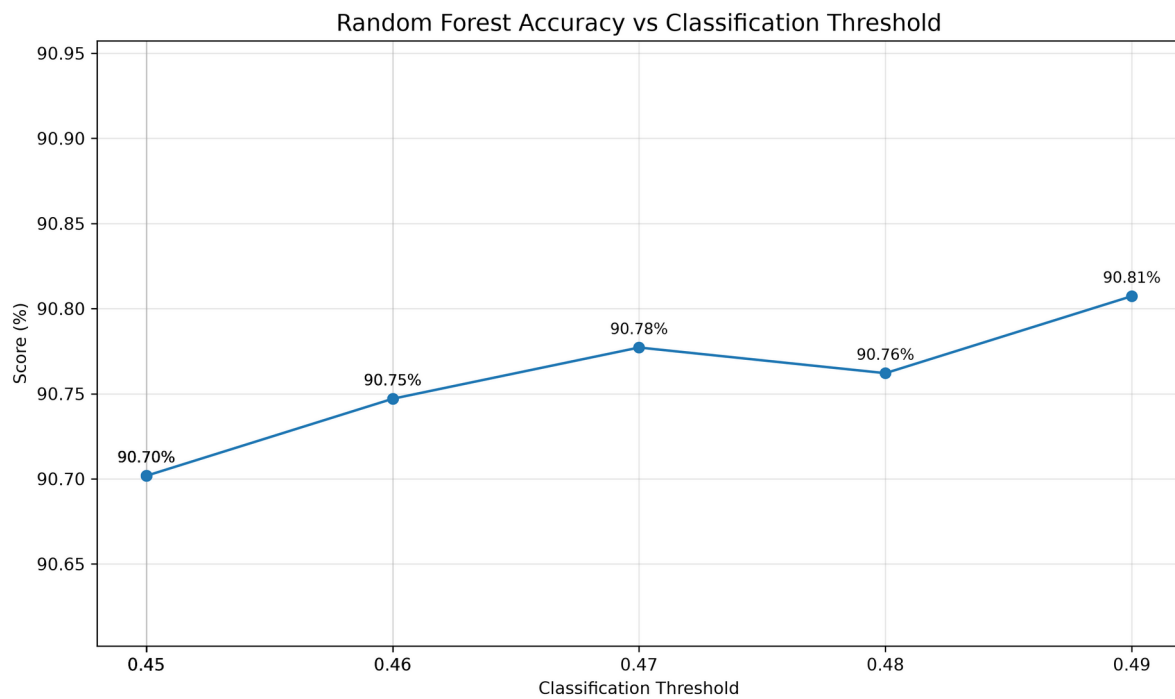
After selecting Random Forest, the project moved from algorithm comparison to controlled parameter optimization. Three aspects were investigated in sequence: classification threshold, number of estimators, and maximum tree depth. The goal was not to maximize every metric independently, because doing so can create conflicting configurations. Instead, the experiments looked for a configuration that preserved strong overall discrimination while increasing recall where possible. Each experiment retained the ten-fold stratified validation methodology, and the graphs were generated with dynamically scaled axes when small changes needed to be made visible. This made subtle performance differences easier to interpret.

9.1 Classification Threshold

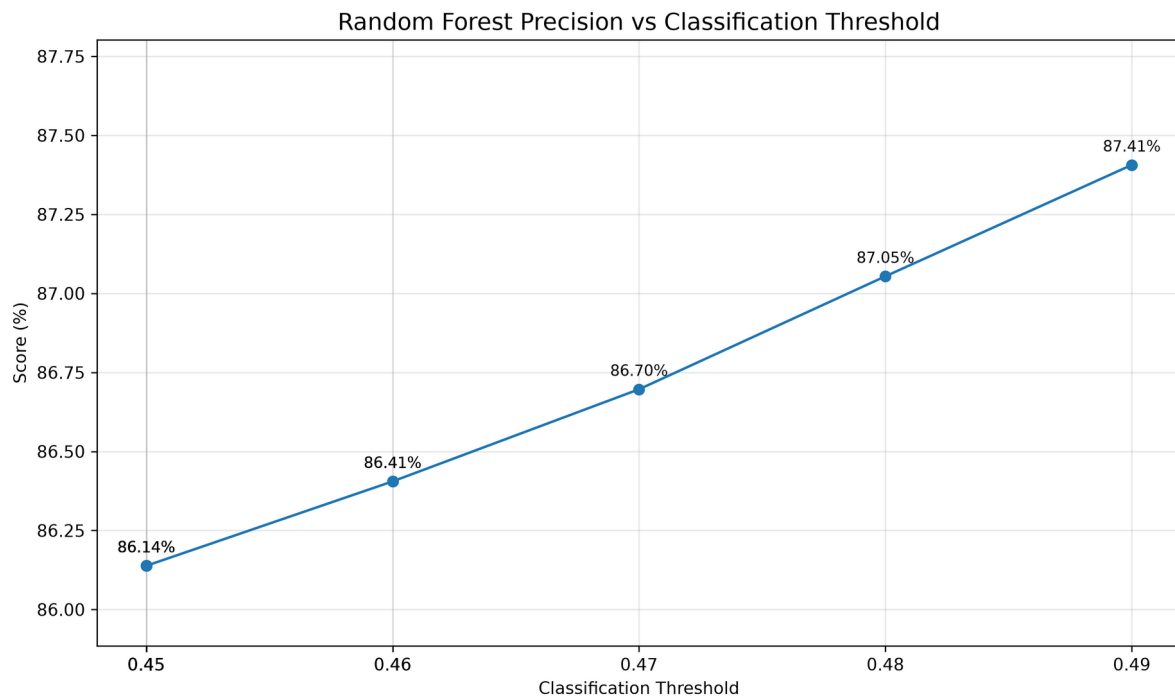
The classification threshold determines when the Random Forest probability output is converted into the positive exoplanet class. The experiment focused on thresholds from 0.45 through 0.49, because earlier testing had already narrowed the useful region. The F1-score graph shows a maximum of 89.13% at thresholds 0.46 and 0.47, compared with 89.11% at 0.45 and lower values at 0.48 and 0.49. The project selected 0.46 rather than moving upward because the application places additional importance on recall. This threshold therefore represents a deliberate operating-point choice rather than an arbitrary deviation from 0.50.

9.1.1 Threshold Experiments

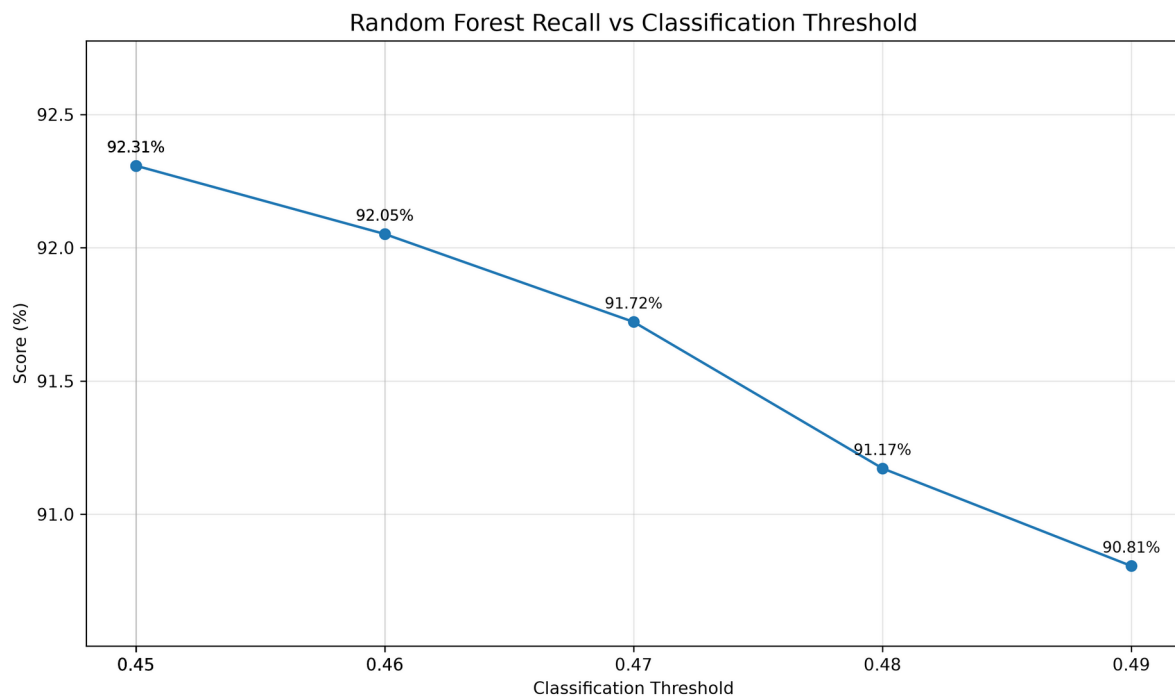
Five threshold experiments were performed at 0.45, 0.46, 0.47, 0.48, and 0.49. The corresponding graphs were generated for accuracy, precision, recall, F1-score, and ROC-AUC. The results show that changing the threshold does not materially alter ROC-AUC because ROC-AUC evaluates the underlying probability ranking, while accuracy, precision, recall, and F1 respond directly to the classification boundary. The narrow performance differences demonstrate that the Random Forest probabilities were already well separated. The selected threshold of 0.46 therefore reflects the preferred balance between recovering positive cases and retaining acceptable precision.



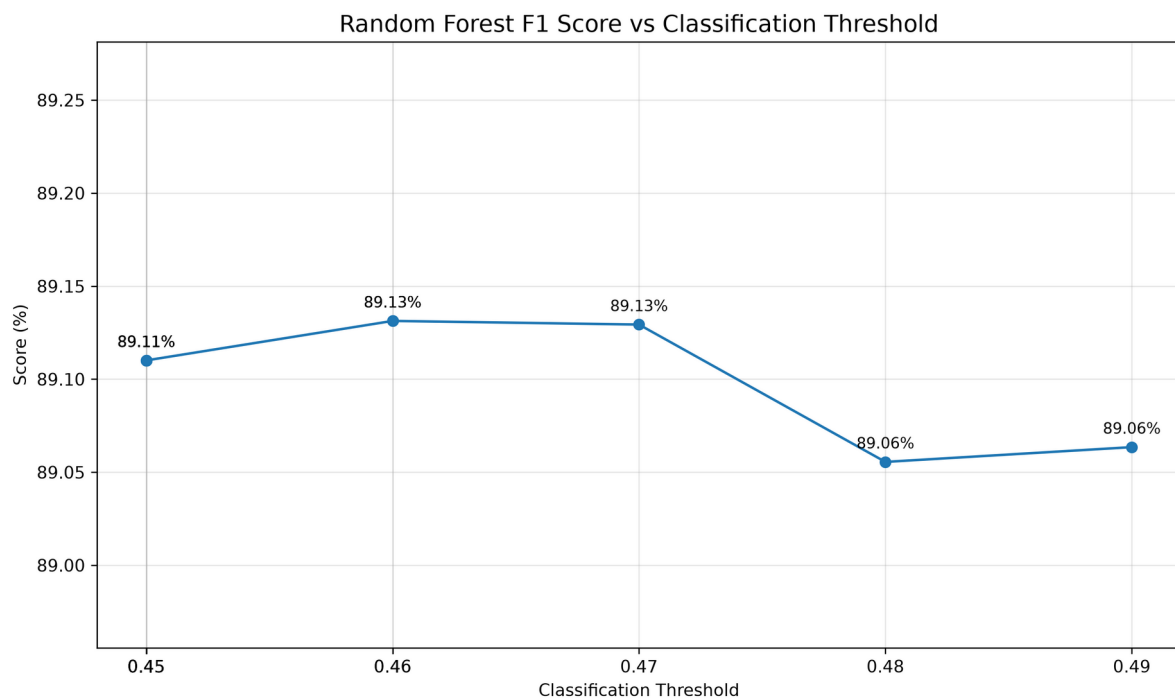
Threshold-tuning result.



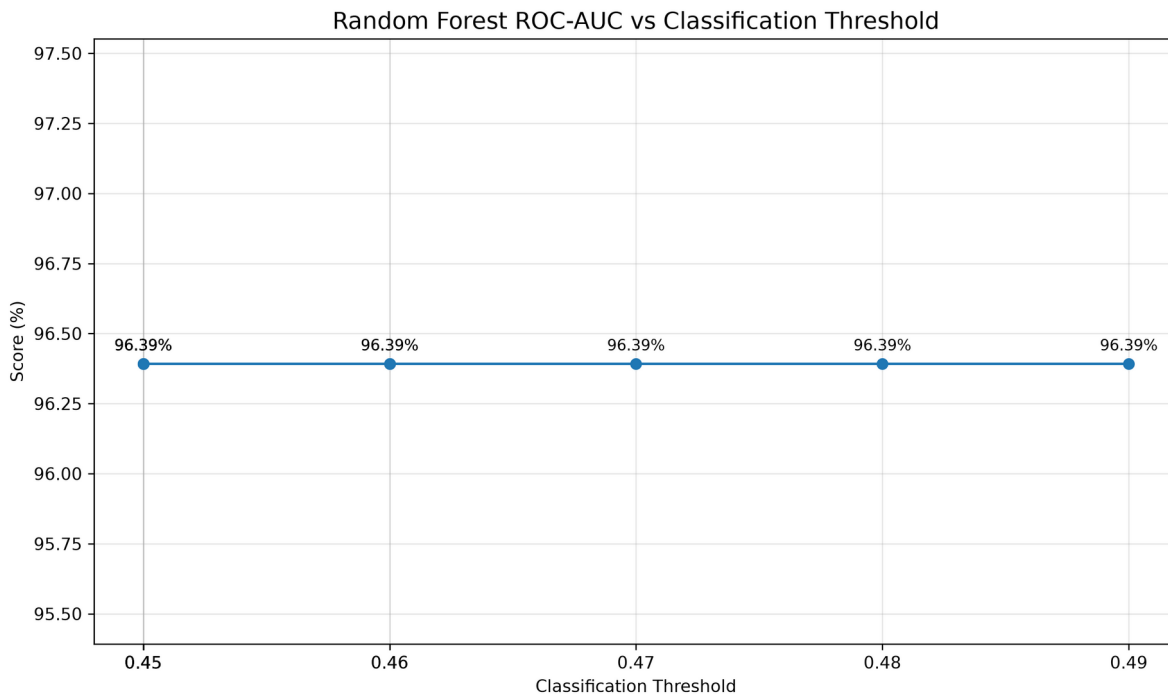
Threshold-tuning result.



Threshold-tuning result.



Threshold-tuning result.



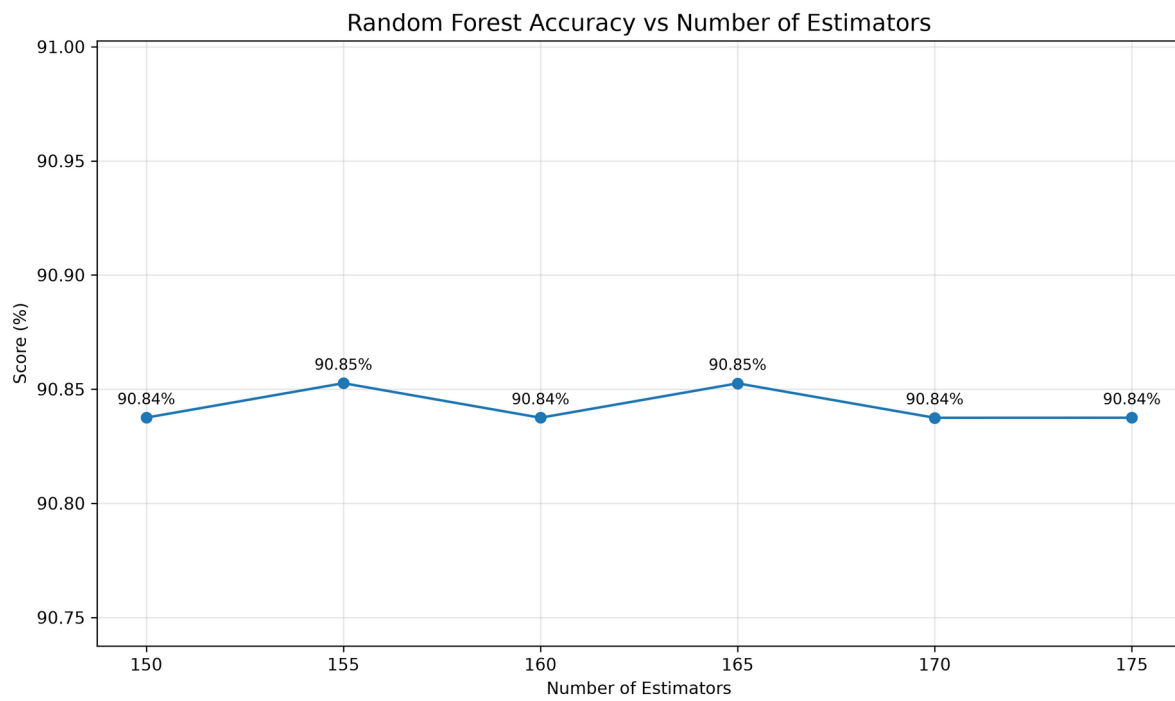
Threshold-tuning result.

9.2 Number of Estimators

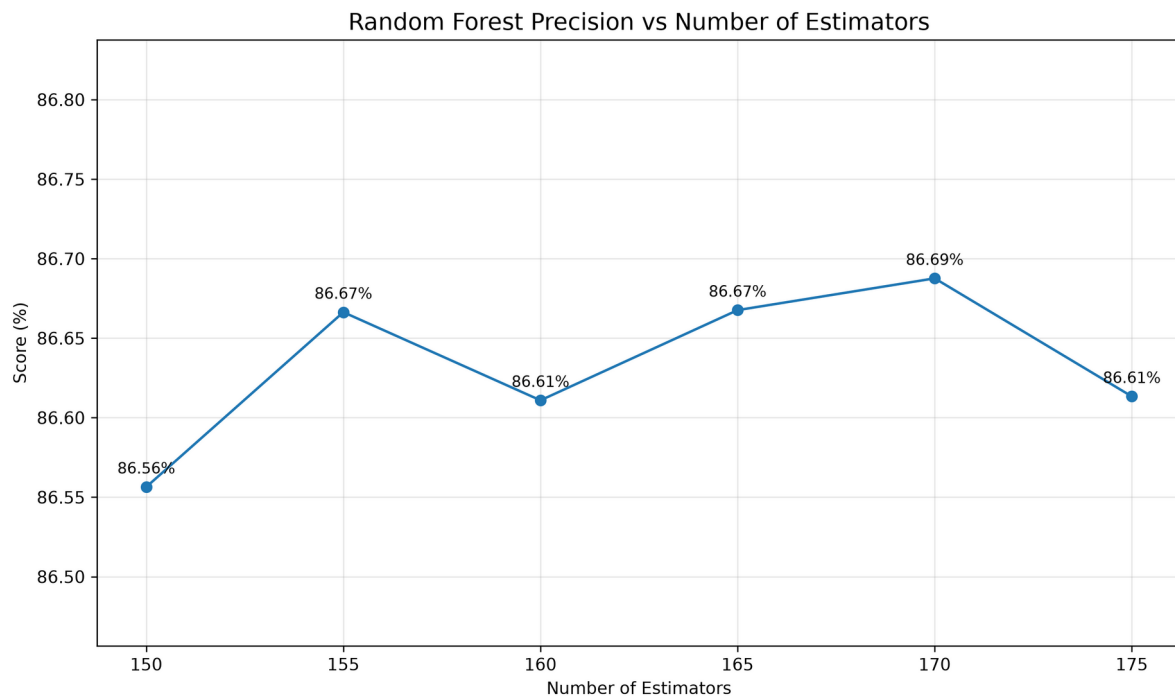
The number of estimators controls how many decision trees participate in the Random Forest. The experiment tested 150, 155, 160, 165, 170, and 175 trees while retaining the selected threshold and other model settings. The F1-score graph shows only very small differences, ranging from approximately 89.21% to 89.24%. The best observed F1-score occurs at 155 and 165 estimators, with 165 selected for the final configuration because it produced a strong balance across the accompanying metrics. The experiment demonstrates diminishing returns: adding more trees beyond this region does not produce a meaningful improvement proportional to the additional model complexity.

9.2.1 Estimator Experiments

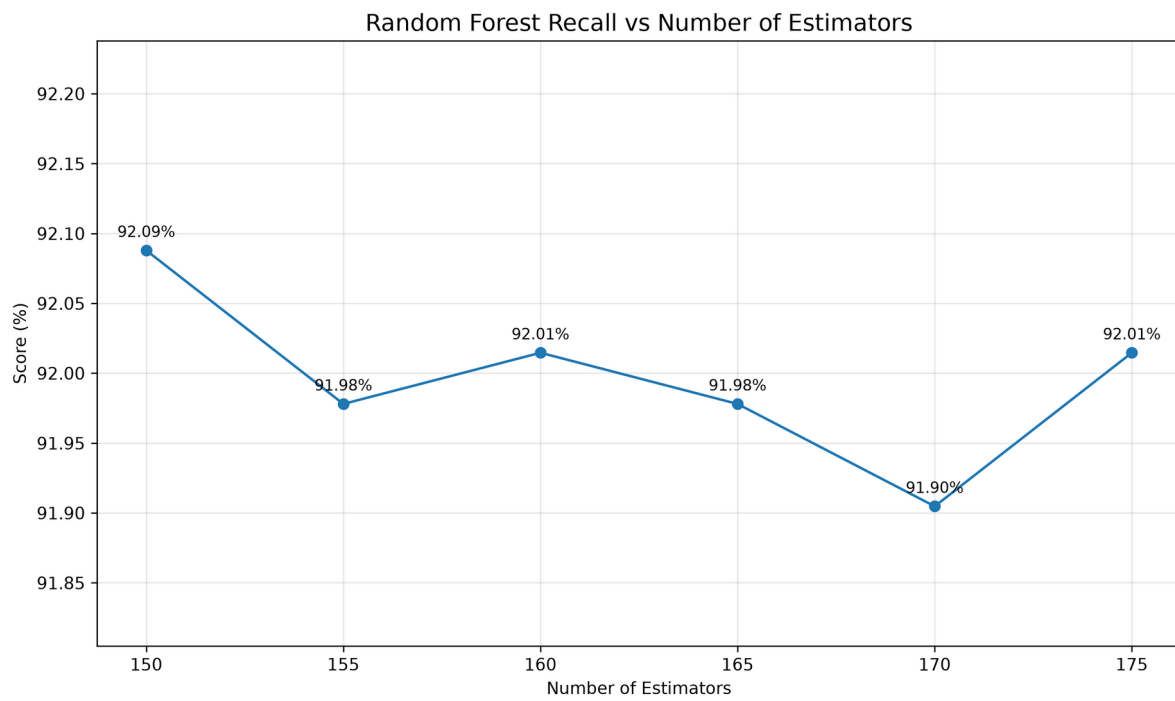
Six estimator configurations were evaluated, and five metric graphs were generated for the experiment: accuracy, precision, recall, F1-score, and ROC-AUC. The curves are deliberately shown with a narrow vertical range because the differences are small and would otherwise appear as an almost flat line. The results indicate that the model is relatively insensitive to moderate changes in the number of trees around the tested range. Since 165 estimators is among the best-performing settings without evidence of a sustained improvement from larger values, it was selected for the final Random Forest rather than continuing to increase the estimator count indefinitely.



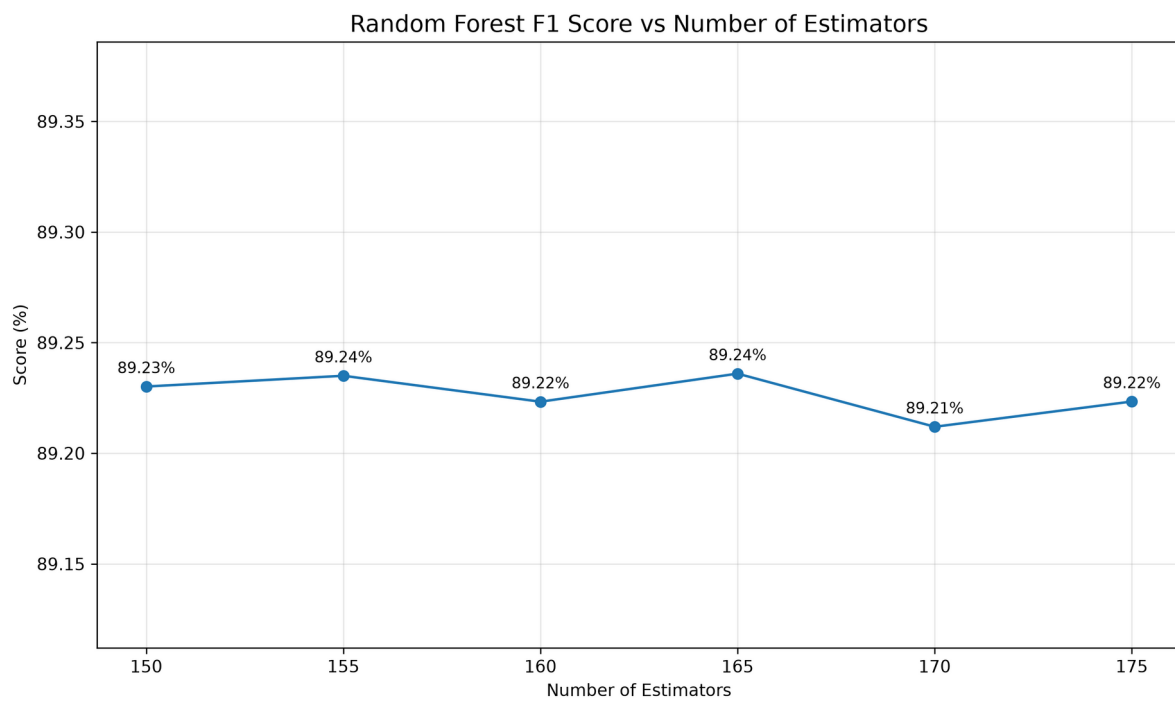
Estimator-count tuning result.



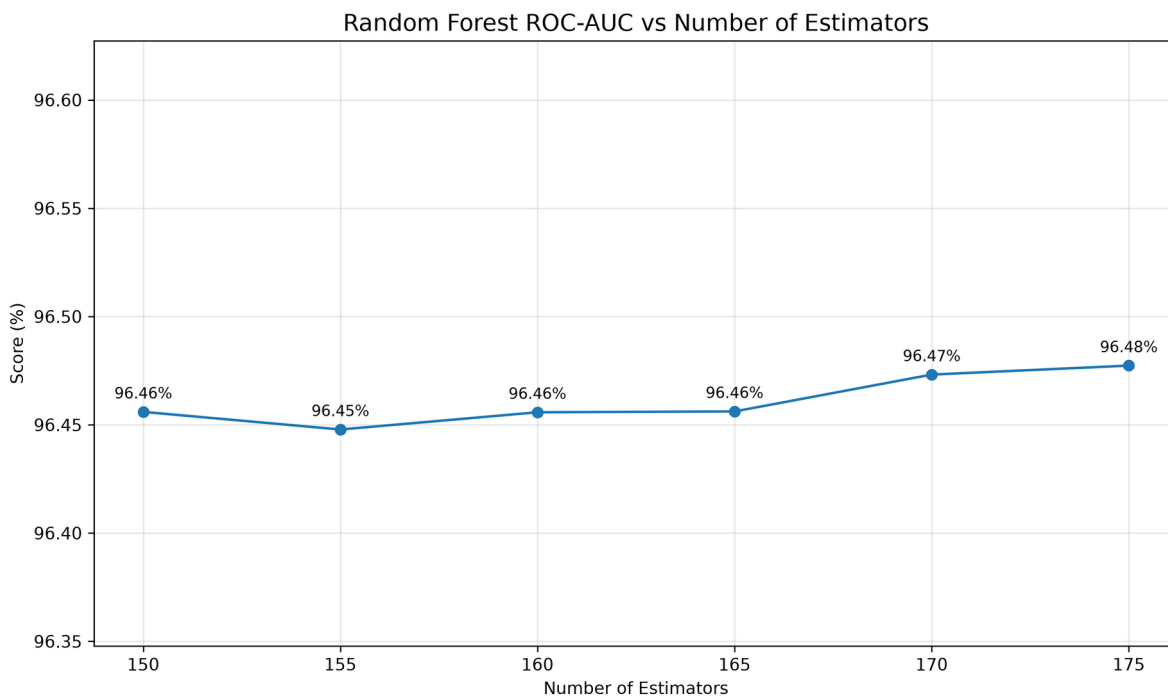
Estimator-count tuning result.



Estimator-count tuning result.



Estimator-count tuning result.



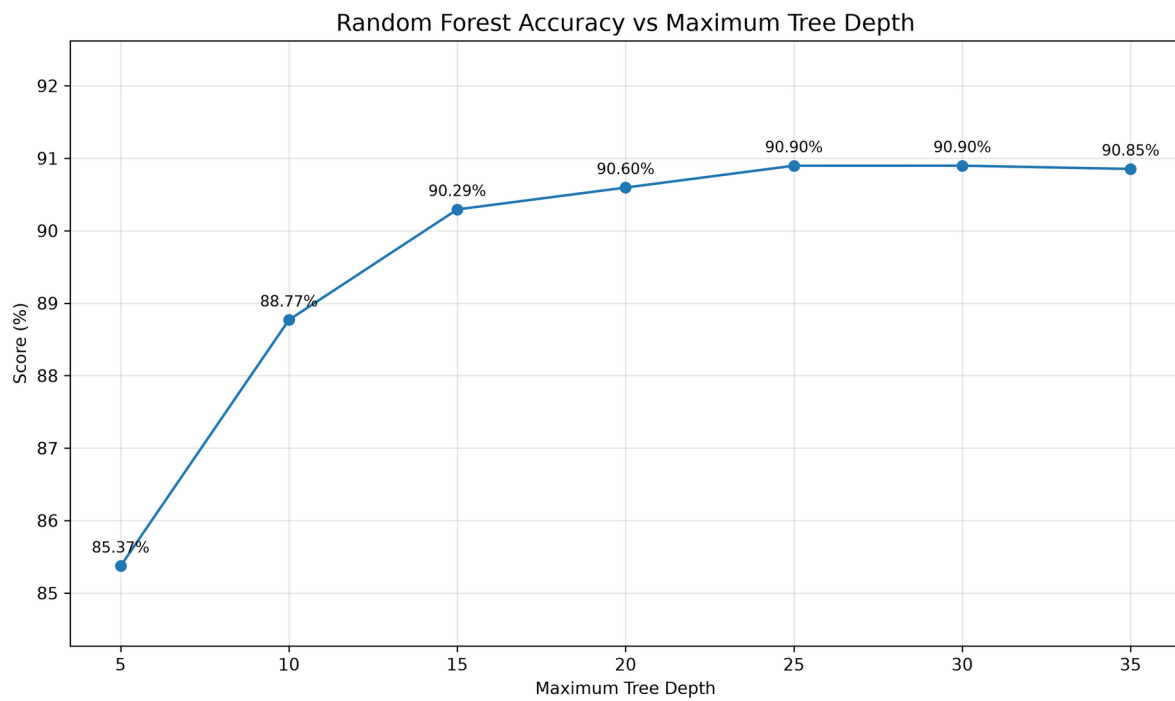
Estimator-count tuning result.

9.3 Maximum Tree Depth

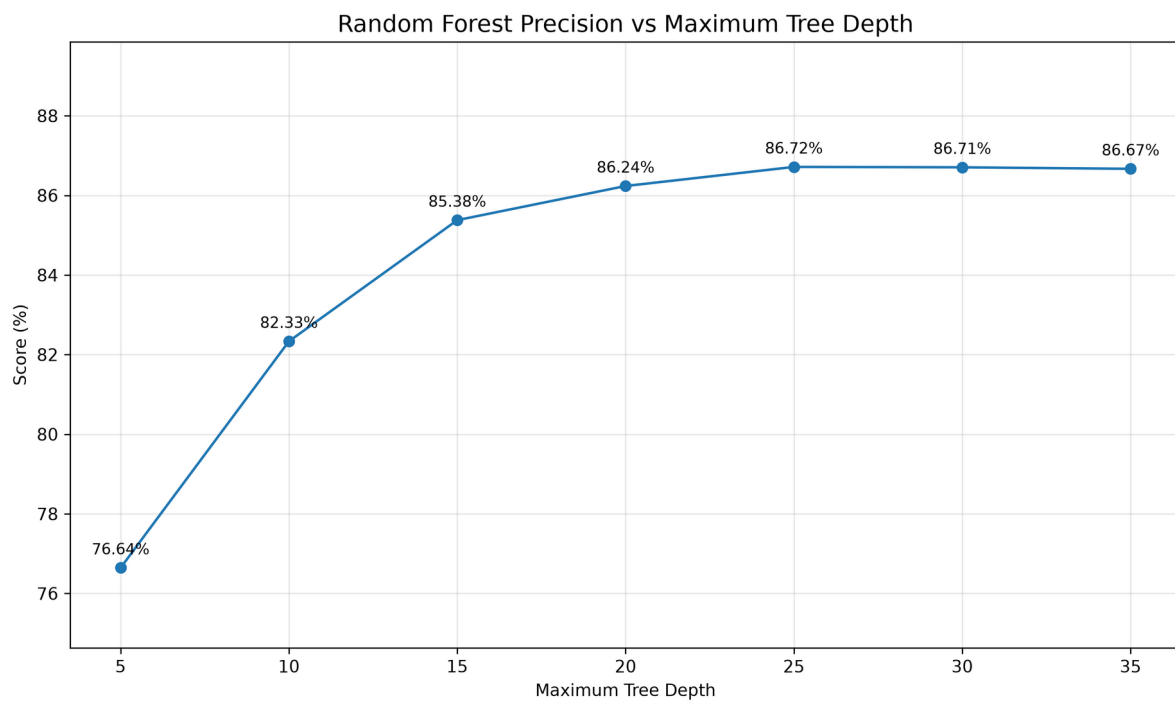
Maximum tree depth controls how deeply individual decision trees are allowed to grow. Seven depth values were tested: 5, 10, 15, 20, 25, 30, and 35. The F1-score rises substantially from 83.96% at depth 5 to 87.19% at depth 10 and continues upward through depths 15 and 20. It reaches 89.29% at both depths 25 and 30 before falling slightly to 89.24% at depth 35. Because depth 30 matches the best observed F1-score while retaining strong results elsewhere, it was chosen for the final configuration after reviewing the complete set of metric graphs.

9.3.1 Depth Experiments

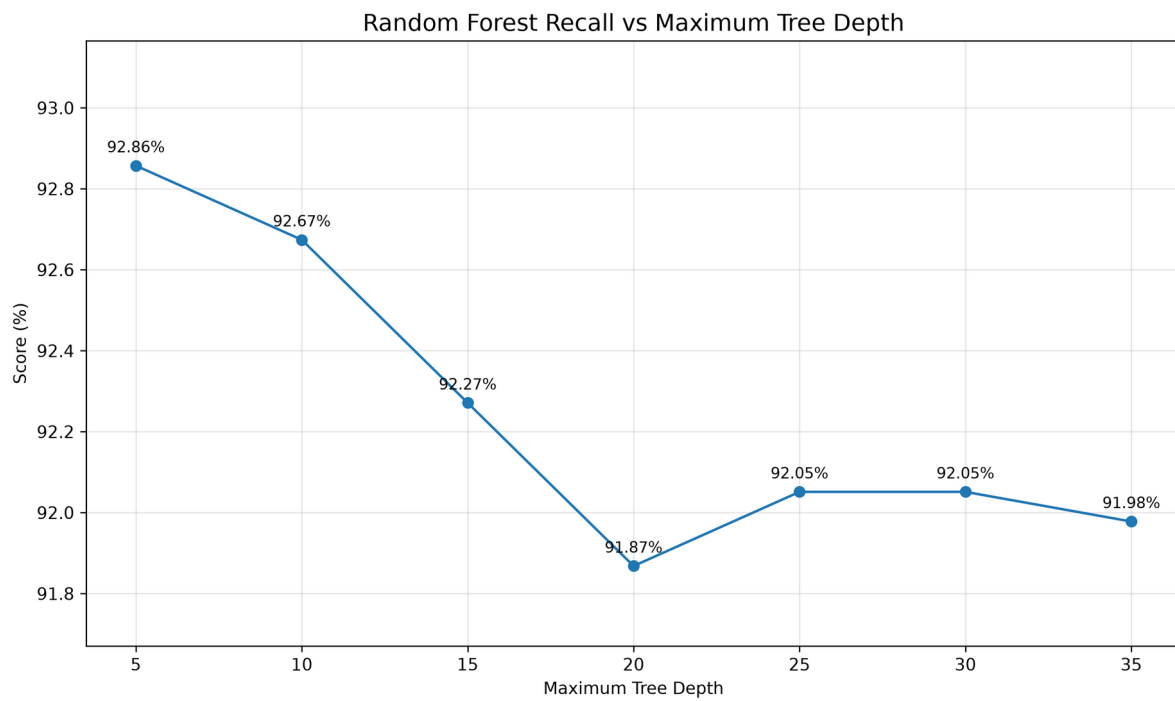
The seven depth experiments provide stronger evidence of a useful tuning effect than the estimator experiment. Shallow trees restrict the model too aggressively and produce noticeably lower F1-score, while increasing depth allows the forest to capture more complex relationships in the numerical features. The improvement begins to level off around depths 25 to 30, indicating diminishing returns from further tree growth. Depth 35 does not improve the result and is slightly worse in F1-score. This plateau supports selecting 30 as the final maximum depth rather than assuming that unrestricted growth must always improve classification.



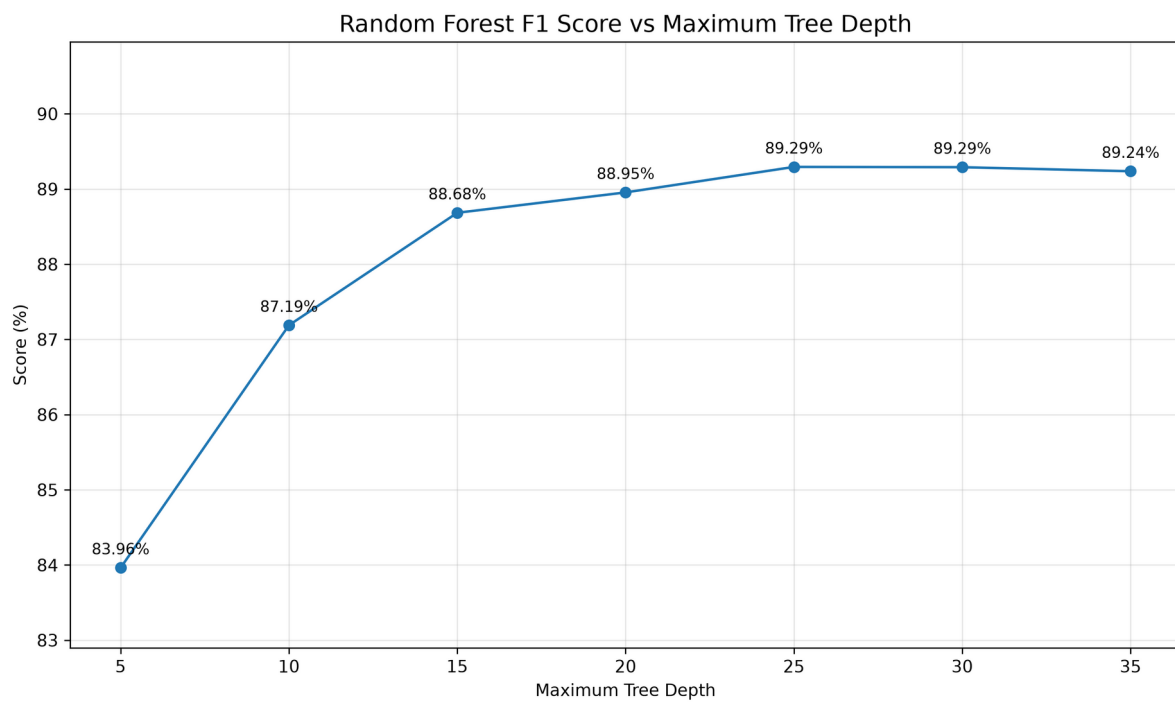
Maximum-depth tuning result.



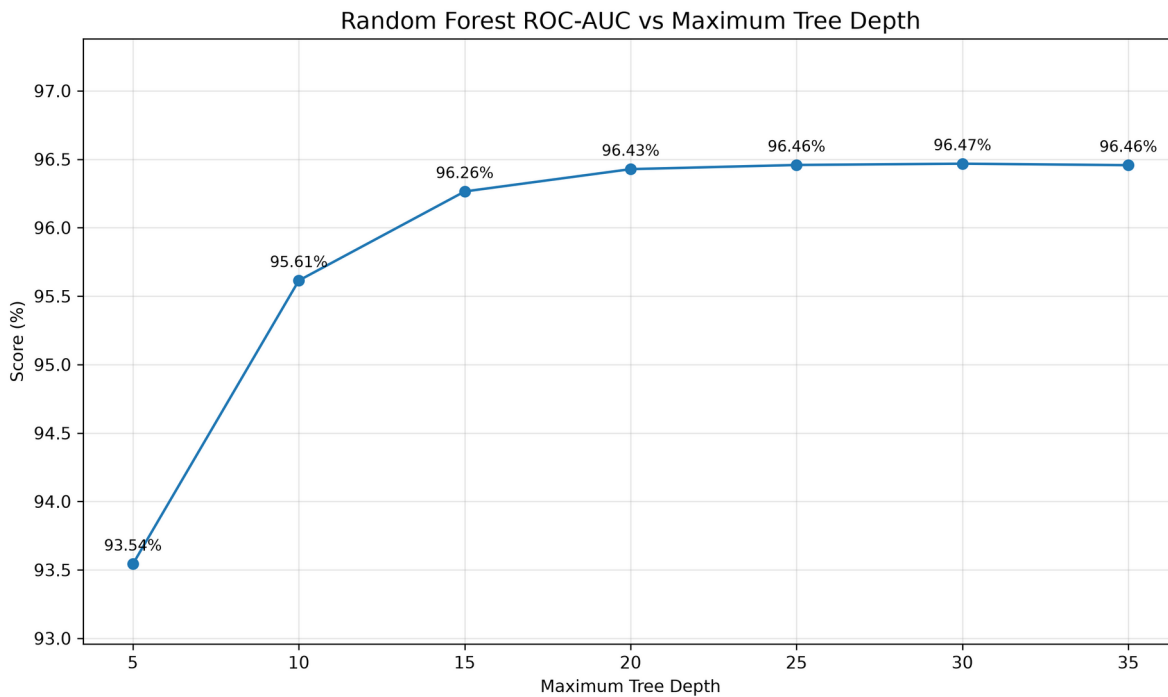
Maximum-depth tuning result.



Maximum-depth tuning result.



Maximum-depth tuning result.



Maximum-depth tuning result.

9.4 Optimization Results

The final Random Forest combines the selected operating threshold and hyperparameters: threshold 0.46, 165 estimators, and maximum depth 30. The baseline Random Forest recorded 90.82% accuracy, 88.03% precision, 90.00% recall, 88.99% F1-score, and 96.39% ROC-AUC in the initial comparison. The final configuration records mean ten-fold values of 90.90% accuracy, 86.71% precision, 92.05% recall, 89.29% F1-score, and 96.47% ROC-AUC. Thus, the optimization produces a small overall improvement while deliberately shifting the balance toward recall.

9.4.1 Why Further Tuning Was Stopped

Further tuning was stopped because the measured improvements had become small and the remaining parameter space would require increasingly fine-grained experimentation for marginal gains. The final configuration improves recall by about two percentage points and slightly improves F1-score and ROC-AUC compared with the baseline, while precision decreases by about 1.32 percentage points. For the intended exoplanet-screening objective, the recall improvement is considered more valuable than preserving every fraction of precision. Continuing to tune additional Random Forest parameters could therefore increase experimental complexity without producing a clearly meaningful practical improvement.

Table 1. Baseline versus final Random Forest performance.

Metric	Baseline RF	Final RF	Change
Accuracy	90.82%	90.90%	+0.08 pp
Precision	88.03%	86.71%	-1.32 pp
Recall	90.00%	92.05%	+2.05 pp
F1-score	88.99%	89.29%	+0.30 pp
ROC-AUC	96.39%	96.47%	+0.08 pp

10. Final Model Evaluation

The final model is a Random Forest configured with 165 estimators, maximum tree depth 30, random_state 42, and a classification threshold of 0.46. Its ten-fold stratified cross-validation results are 90.90% accuracy, 86.71% precision, 92.05% recall, 89.29% F1-score, and 96.47% ROC-AUC. The final visualizations are generated from out-of-fold predictions, meaning each observation is predicted by a fold model that did not use that observation for training. This avoids the training-set leakage that would otherwise produce artificially perfect-looking final graphs.

10.1 Confusion Matrix

The final out-of-fold confusion matrix contains 3,509 correctly classified non-exoplanet observations and 2,513 correctly classified exoplanet observations. There are 386 false positives and 217 false negatives. The relatively small number of false negatives is important for the project's objective because missed exoplanet candidates are considered more costly than accepting some additional false positives. The threshold of 0.46 contributes to this behavior by making the classifier slightly more willing to assign the positive class than a conventional 0.50 threshold would. The matrix therefore gives a concrete view of the precision-recall trade-off behind the final configuration.

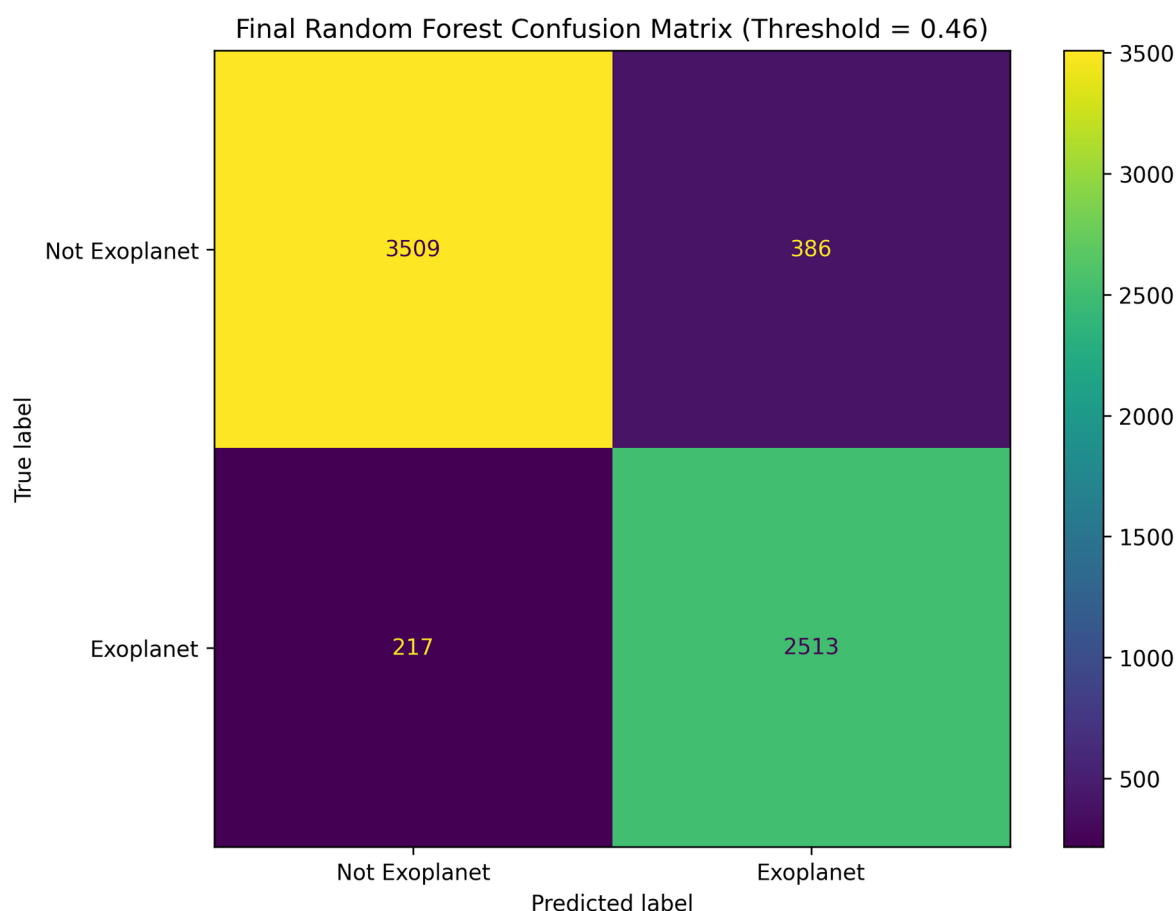


Figure 21. Final out-of-fold confusion matrix at threshold 0.46.

10.2 ROC Curve

The final ROC curve is generated from the out-of-fold probability predictions rather than from hard class labels. This is important because a ROC curve evaluates the model over a range of possible decision thresholds. The resulting ROC-AUC is approximately 0.9647, consistent with the mean ten-fold ROC-AUC reported in the performance record. The curve demonstrates strong separation between the two classes, while the diagonal reference line represents random classification. The selected threshold of 0.46 is an operating point on this broader probability-ranking behavior rather than a parameter that defines the ROC curve itself.

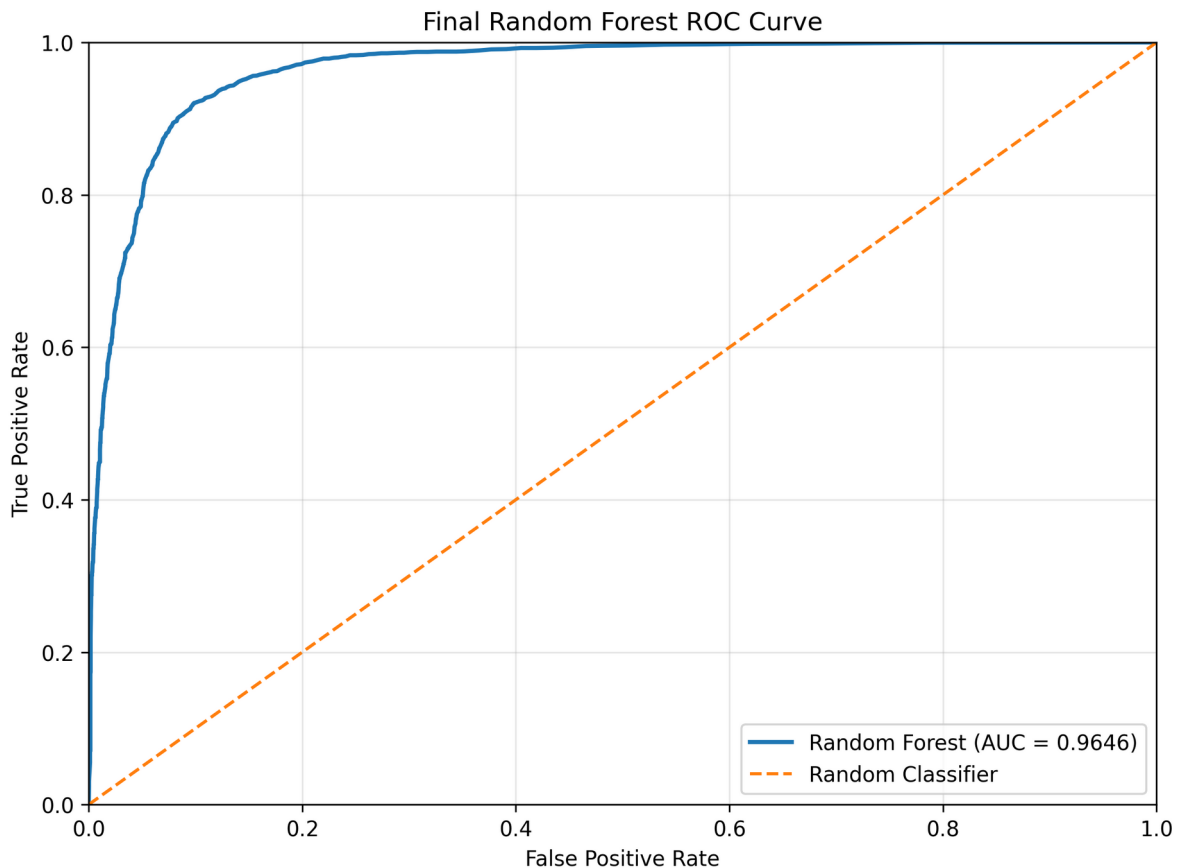


Figure 22. Final out-of-fold ROC curve.

10.3 Final Metrics

The final ten-fold results show a balanced and practically useful classifier. Accuracy is 90.90%, meaning that roughly nine out of ten evaluated observations are classified correctly. Precision is 86.71%, while recall reaches 92.05%, indicating that the model successfully recovers most actual exoplanet cases. The resulting F1-score is 89.29%, and ROC-AUC is 96.47%, showing strong overall ranking performance. The difference between precision and recall is intentional: the project accepts somewhat more false positives in exchange for reducing false negatives, which aligns with the stated objective of minimizing the risk of overlooking genuine planetary candidates.

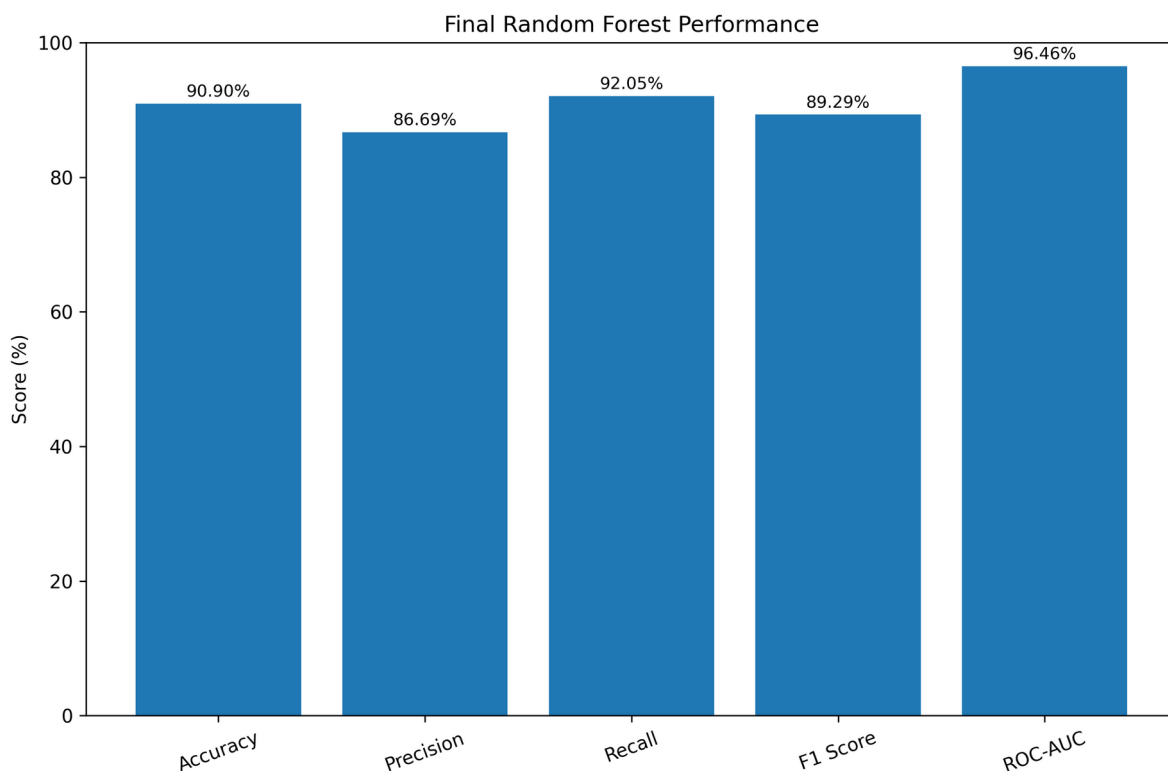


Figure 23. Final out-of-fold performance metrics.

10.4 Interpretation

The final evaluation suggests that the model has learned a strong relationship between the selected KOI numerical measurements and the binary exoplanet target. The high ROC-AUC indicates that the probability scores generally distinguish the classes effectively, while the 92.05% recall demonstrates that the selected operating threshold captures a large majority of positive cases. At the same time, the precision value shows that the classifier is not perfect and should be regarded as a screening aid rather than an independent astronomical confirmation mechanism. A prediction from the model should therefore support prioritization and analysis rather than replace scientific validation.

11. Conclusion

This project developed and evaluated a complete machine-learning workflow for binary classification of Kepler Objects of Interest. Starting with nine selected numerical features, nine supervised algorithms were compared using a common ten-fold stratified cross-validation procedure. Random Forest was selected because it provided the strongest overall combination of accuracy, F1-score, precision, and ROC-AUC in the initial tests. Controlled optimization then established a classification threshold of 0.46, 165 estimators, and maximum depth 30. The final model reached 90.90% accuracy, 92.05% recall, 89.29% F1-score, and 96.47% ROC-AUC, providing a strong experimental result while maintaining an emphasis on detecting positive candidates.

12. Future Improvements

Several improvements could be investigated in future work without changing the core architecture established here. Hyperparameter optimization could be expanded to parameters such as minimum samples per split, minimum samples per leaf, maximum

features, class weighting, and bootstrap behavior. Calibration could also be examined so that predicted probabilities have a more explicit probabilistic interpretation. Additional validation on a temporally separated or independently sourced dataset would provide a stronger test of generalization. Finally, feature-importance analysis, SHAP-style explanations, or comparison with additional modern classifiers could help determine which astronomical measurements contribute most strongly to the model's decisions.

13. References

The report uses the NASA Exoplanet Archive as the primary data source and the project's GitHub repository as the software and implementation reference. The references are formatted in a numbered, AMA-style presentation so that sources can be identified consistently in the report. The NASA source should be cited for the cumulative KOI dataset used to construct the experiments, while the repository provides the implementation context, including preprocessing, model training, tuning, and visualization programs. Software libraries used in the project are also listed to make the computational environment and methodological dependencies clear.

13.1 NASA Dataset

The NASA Exoplanet Archive provides the cumulative exoplanet and Kepler Objects of Interest tables used as the source of the numerical observations in this project. The dataset was accessed for the project through the archive's cumulative table interface. In a reproducible research context, the archive should be treated as the authoritative source for the downloaded observations, while the processed CSV used by the classifier represents the project's derived working dataset. The reference below records the organization, resource title, access date, and direct archive address in a numbered bibliographic style.

[1] NASA Exoplanet Archive. Cumulative Exoplanet Data. California Institute of Technology. Accessed September 15, 2026.

<https://exoplanetarchive.ipac.caltech.edu/cgi-bin/TblView/nph-tblView?app=ExoTbls&config=cumulative>

13.2 GitHub Repository

The project source code is maintained in a GitHub repository containing the implementation developed during the experimental process. The repository provides the software context for the report, including preprocessing, model-training logic, Random Forest tuning, and visualization generation. Referencing the repository is useful because the numerical results in this report are connected to executable code rather than being presented as isolated calculations. The repository link should be retained with the final submission so that the implementation and the report can be considered together when the project is evaluated.

[2] Likhith2062. ExoPlanetDetection. GitHub. Accessed September 15, 2026.

<https://github.com/Likhith2062/ExoPlanetDetection>

13.3 Libraries / Relevant Software

The implementation relies primarily on Python and established scientific-computing libraries. Pandas is used for tabular data handling, scikit-learn supplies the machine-learning estimators, cross-validation procedures, and evaluation metrics, Joblib

supports parallel execution of validation folds, Matplotlib is used to create report-ready graphs, and Joblib serialization is used to save trained models. These libraries were selected because they provide transparent, widely used implementations of the algorithms required by the project. Their use also keeps the experimental workflow reproducible without introducing unnecessary software complexity.

[3] Pedregosa F, et al. Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research. 2011;12:2825-2830.

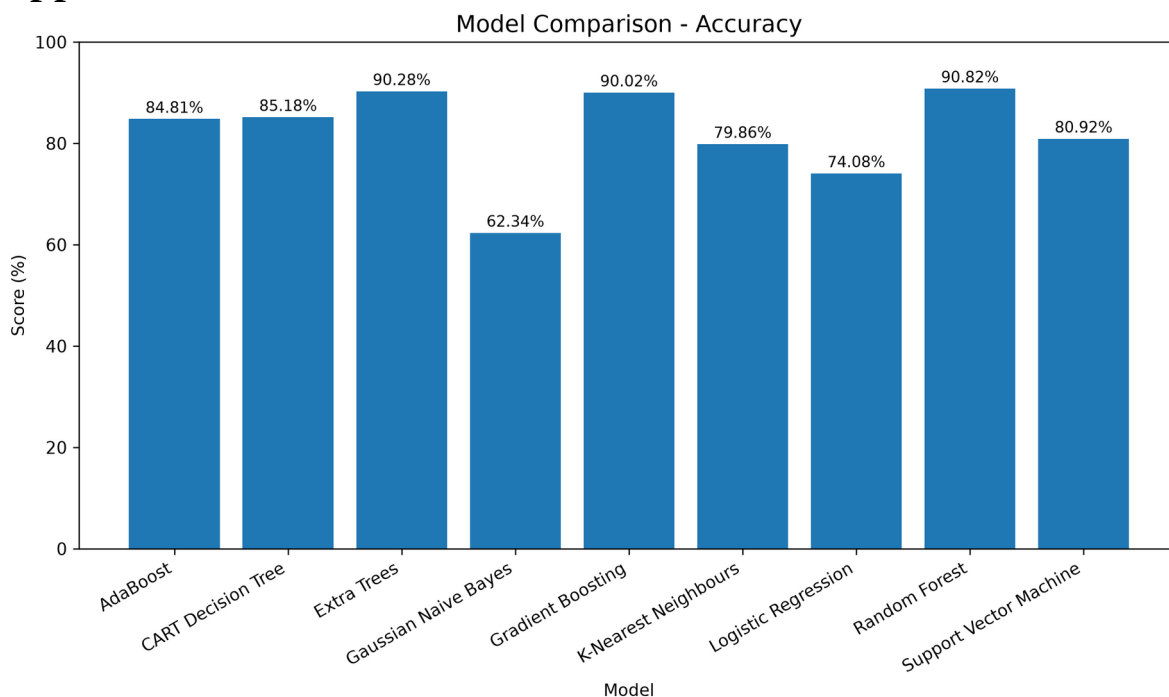
[4] McKinney W. Python for Data Analysis. 3rd ed. O'Reilly Media; 2022.

[5] Hunter JD. Matplotlib: A 2D Graphics Environment. Computing in Science & Engineering. 2007;9(3):90-95.

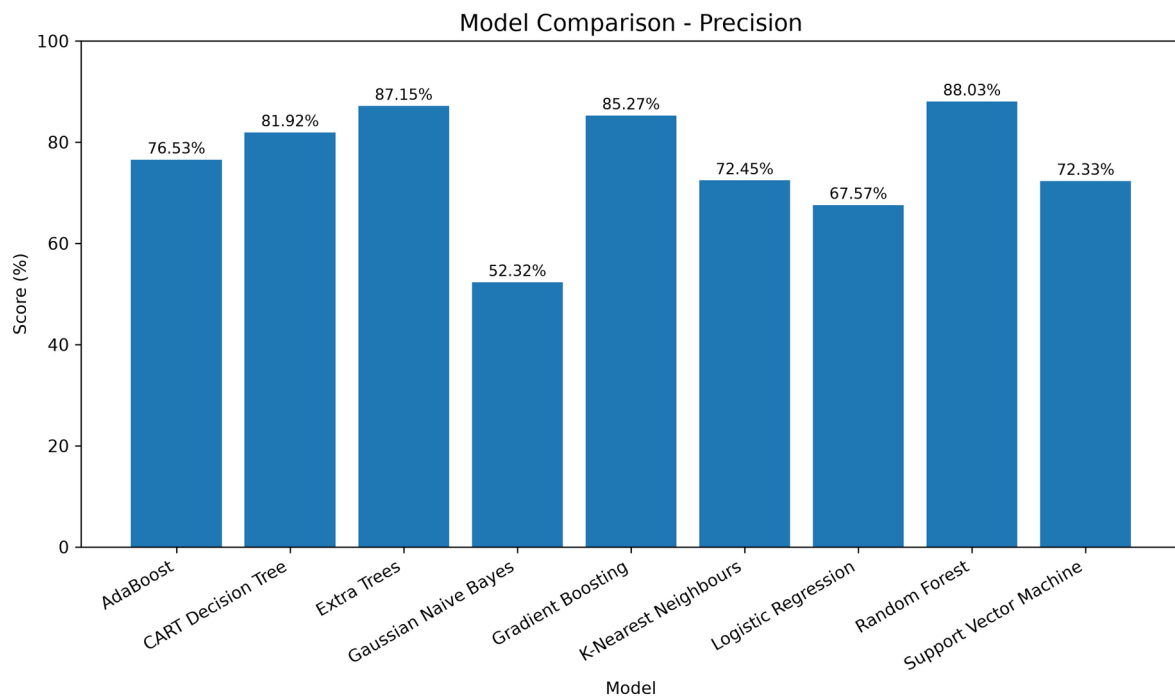
14. Appendix - All Graphs Generated During the Experiments

This appendix collects the graphical outputs produced during the experimental stages of the project. The main report places the most relevant graphs next to the decisions they support, while this appendix preserves the complete set of generated metric plots for reference. The graphs include the nine-model initial comparison, all threshold-tuning metrics, all estimator-tuning metrics, all depth-tuning metrics, and the final Random Forest evaluation outputs. Keeping the complete graphical record makes the optimization process auditable and allows a reader to inspect trends that may not be fully described in the narrative sections.

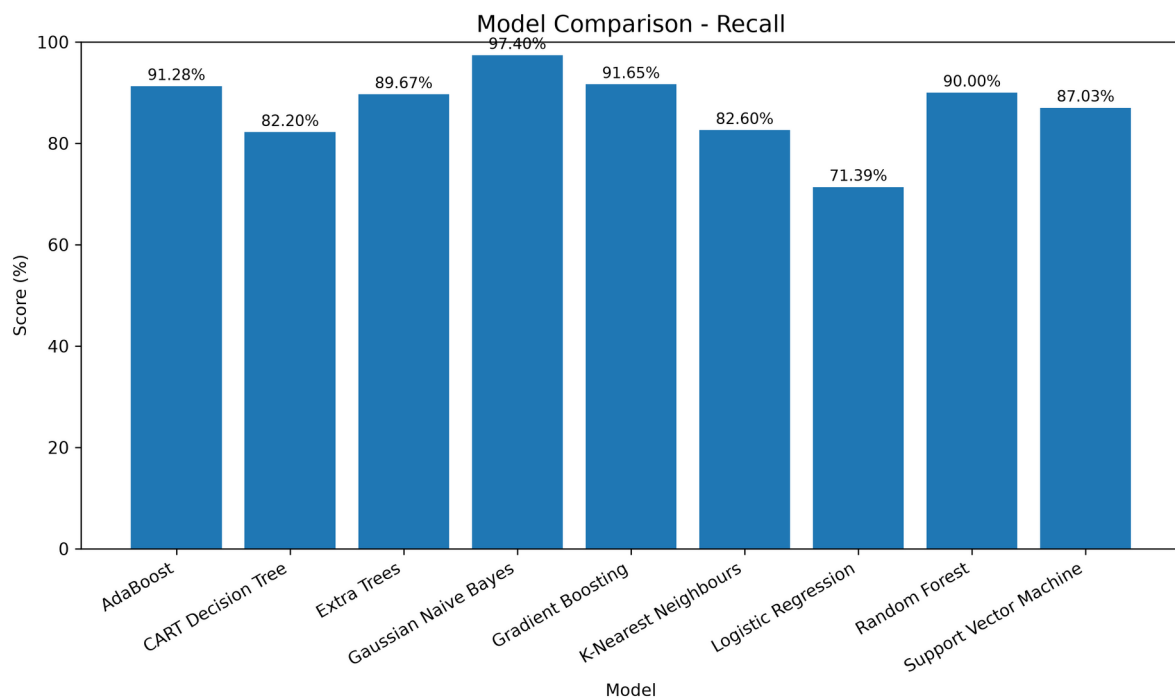
Appendix A - New Architecture Initial Tests



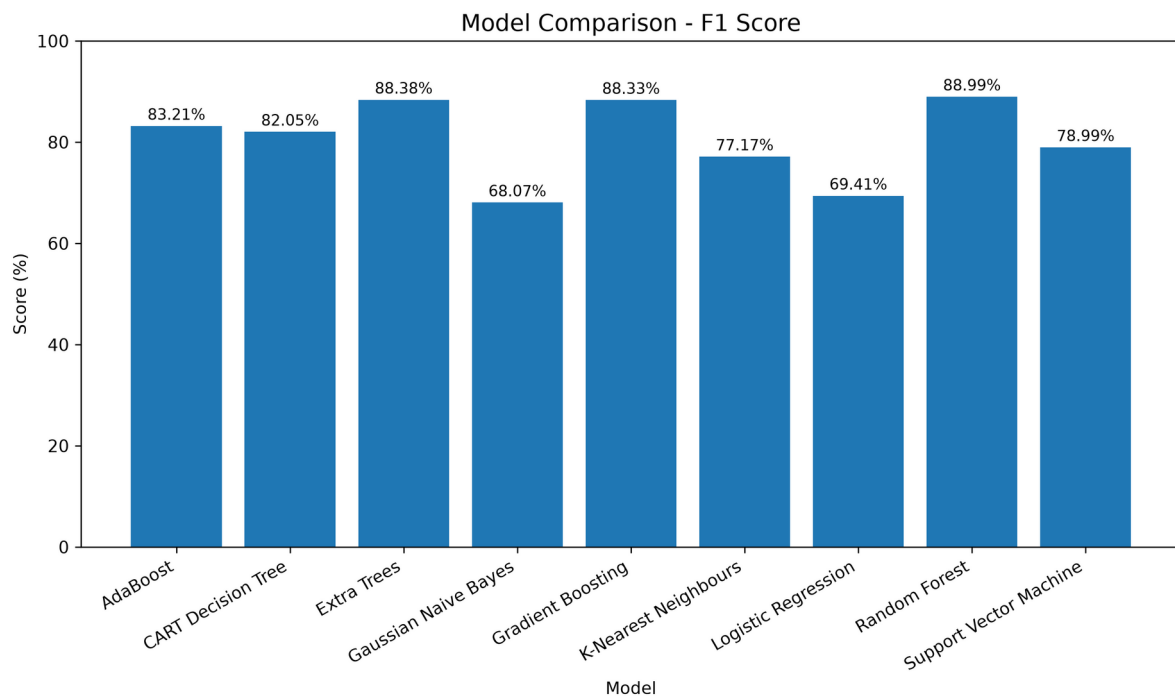
Appendix graph: Model Accuracy.



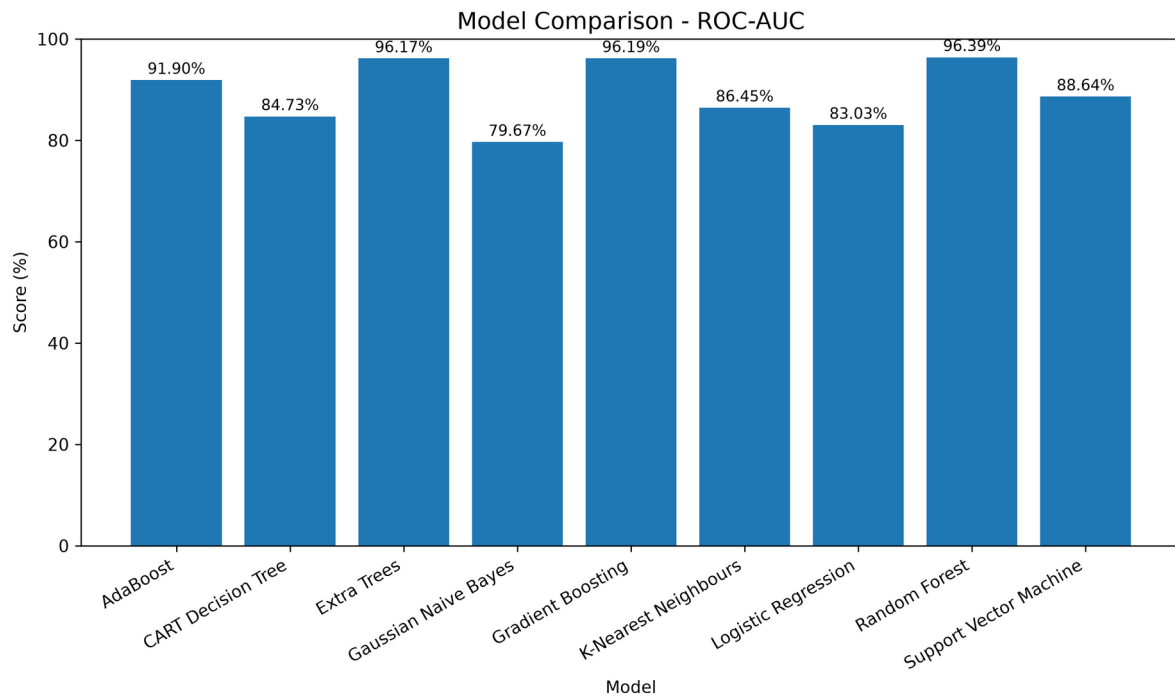
Appendix graph: Model Precision.



Appendix graph: Model Recall.

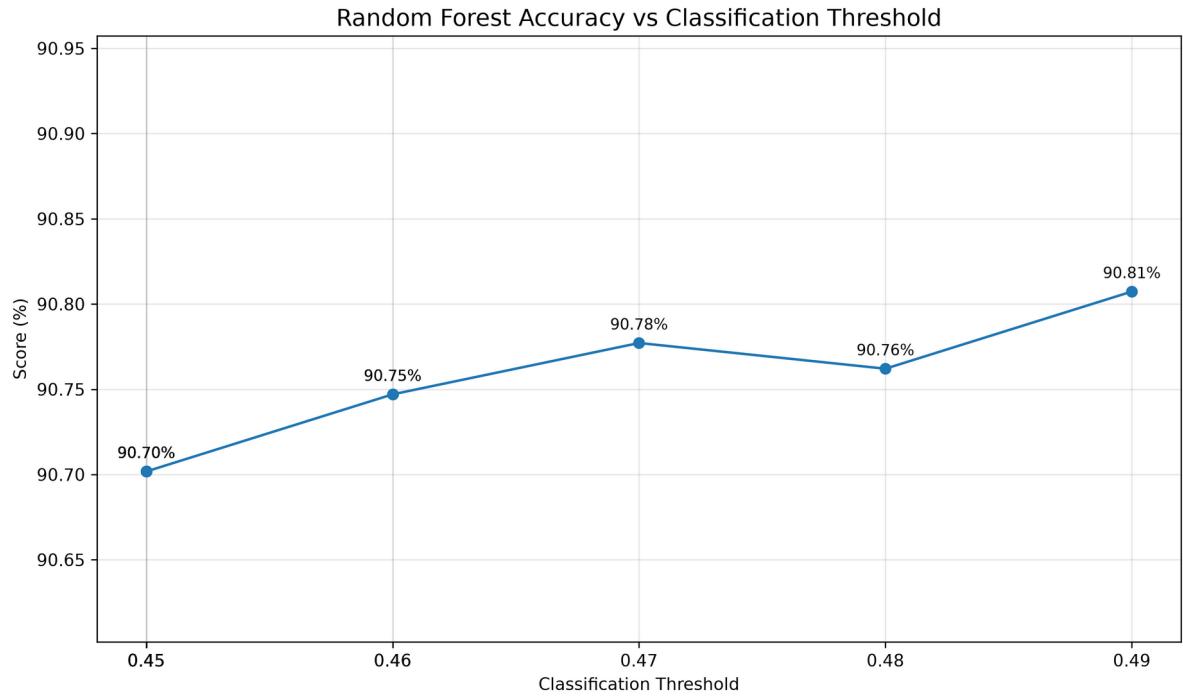


Appendix graph: Model F1.

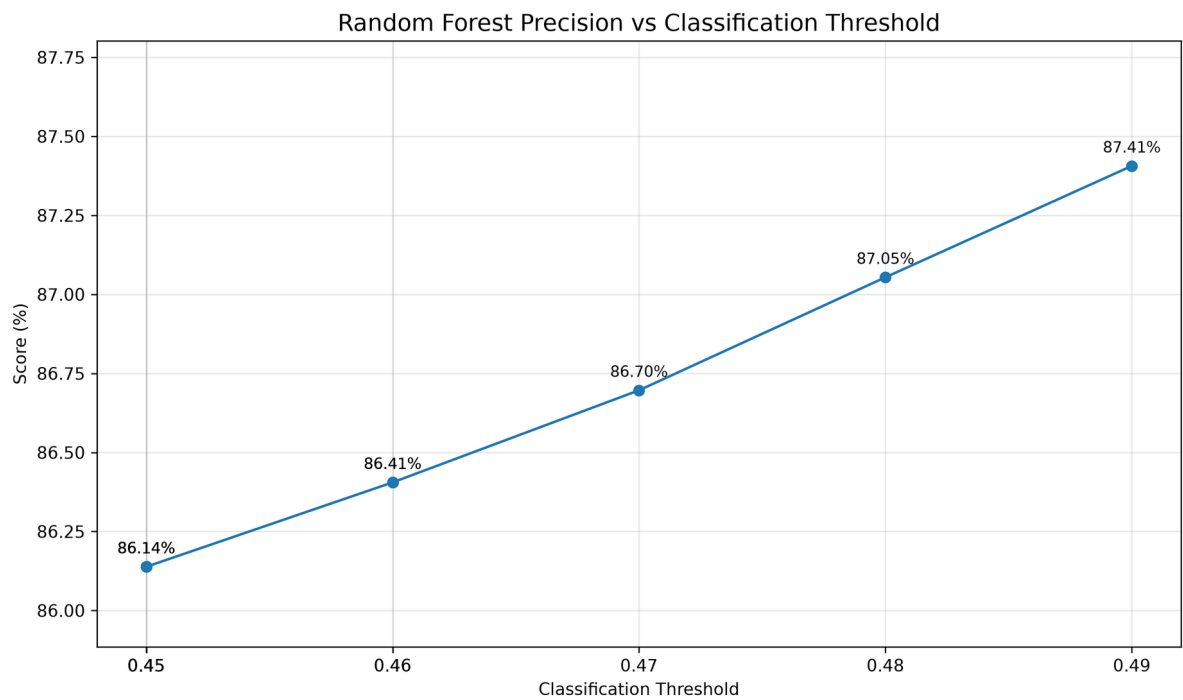


Appendix graph: Model Roc Auc.

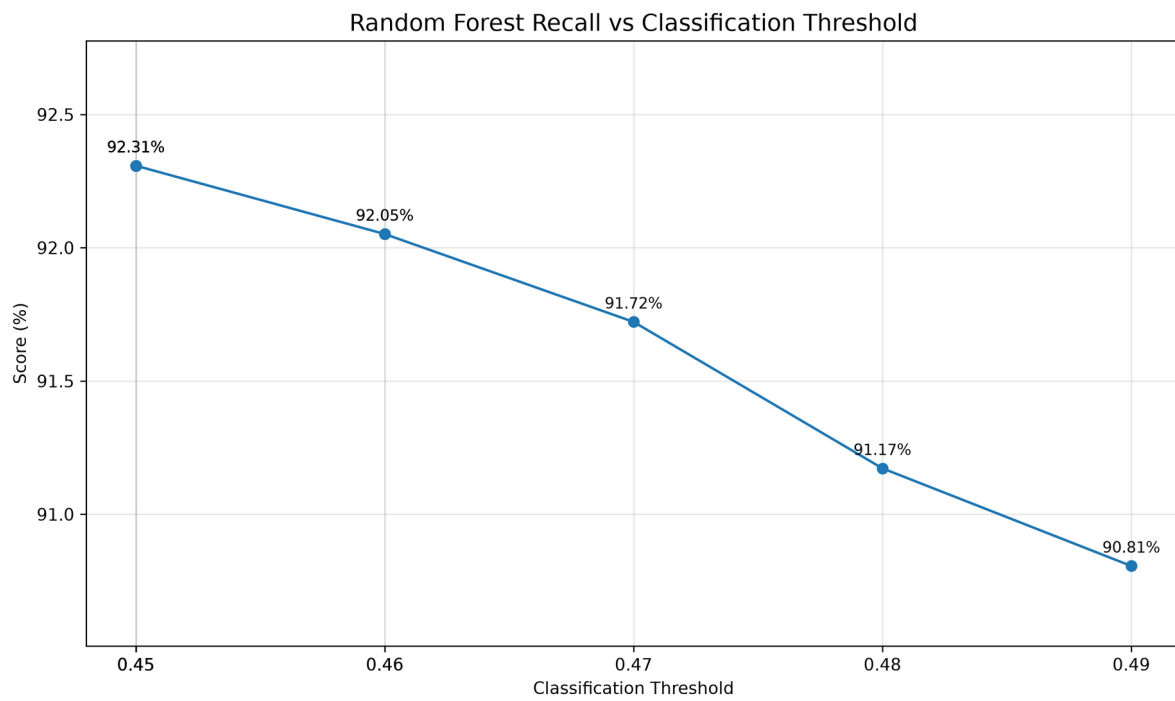
Appendix B - Threshold Tuning Tests



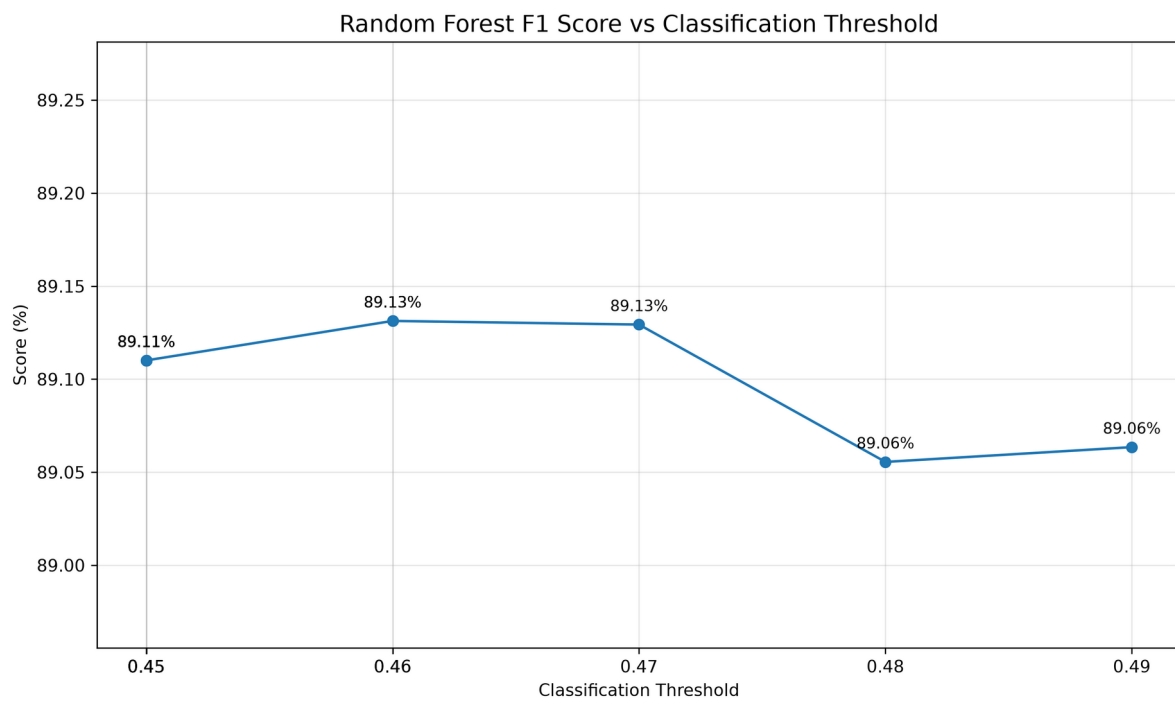
Appendix graph: Threshold Accuracy.



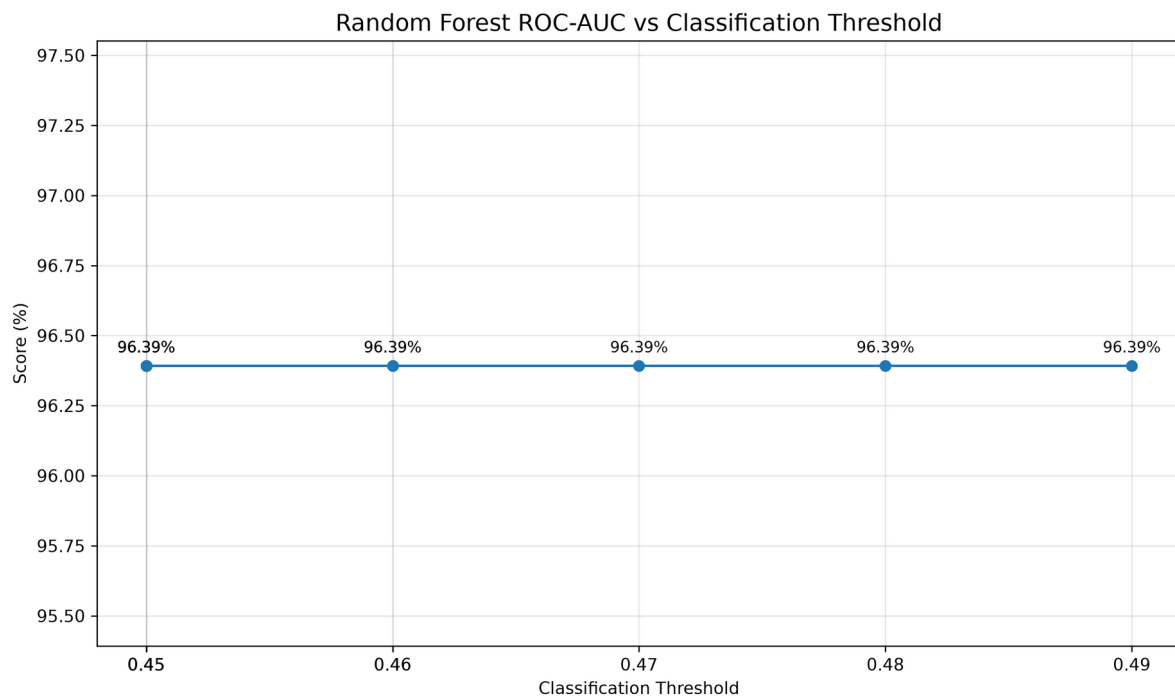
Appendix graph: Threshold Precision.



Appendix graph: Threshold Recall.

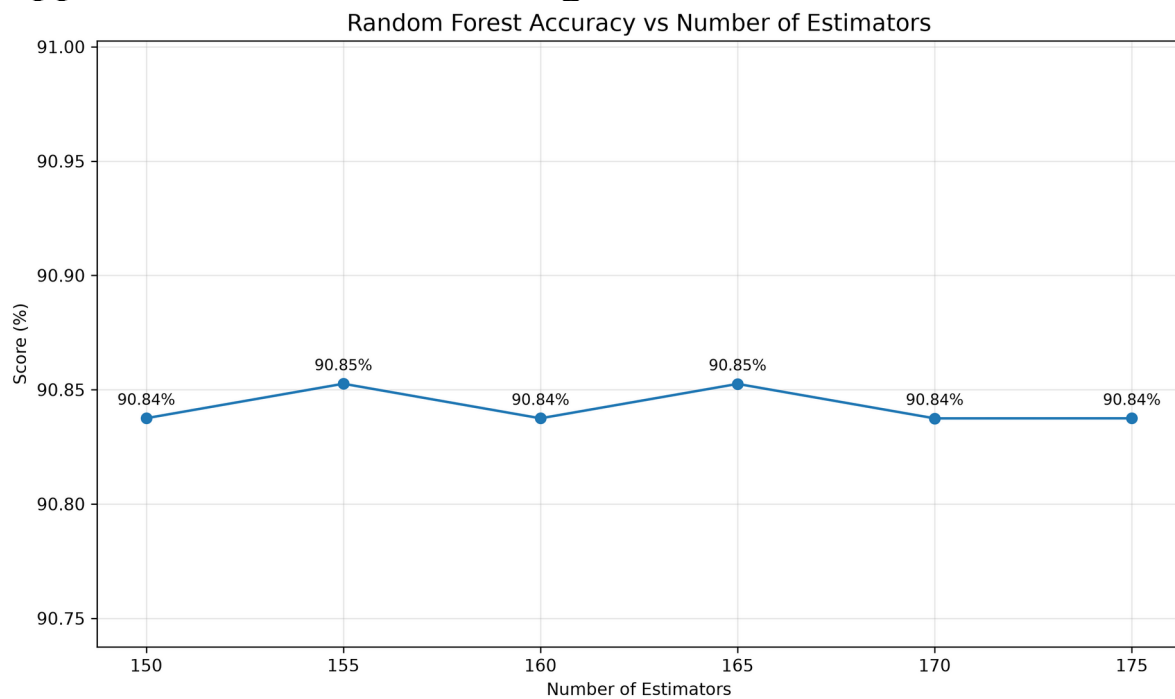


Appendix graph: Threshold F1.

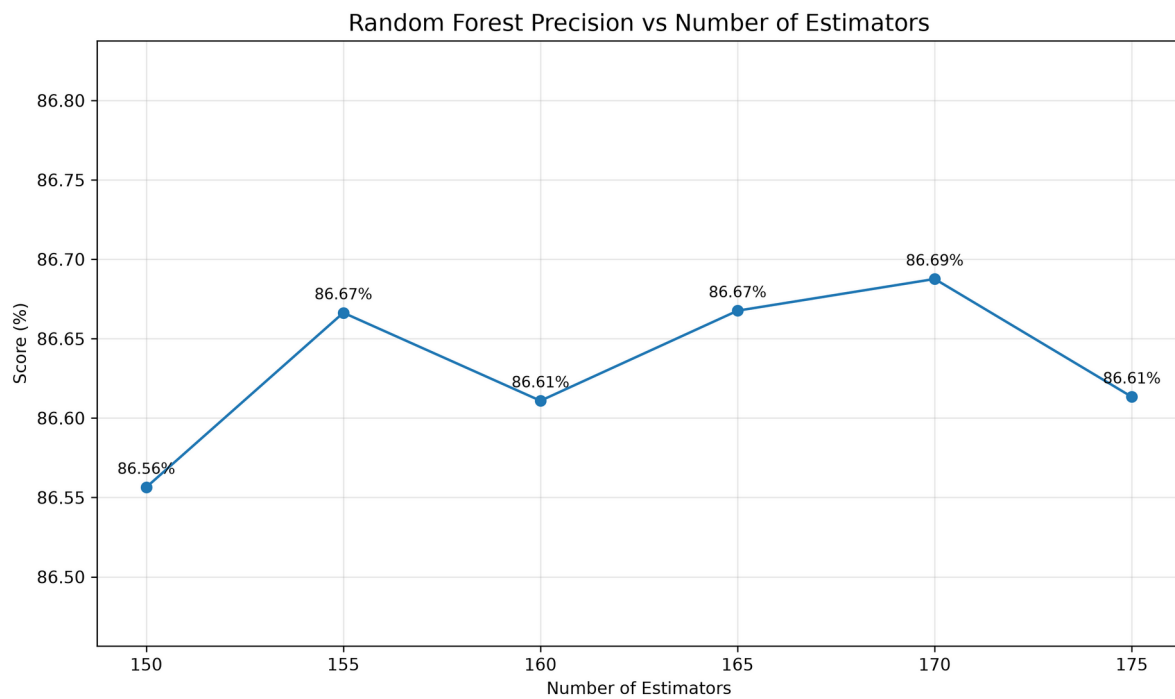


Appendix graph: Threshold Roc Auc.

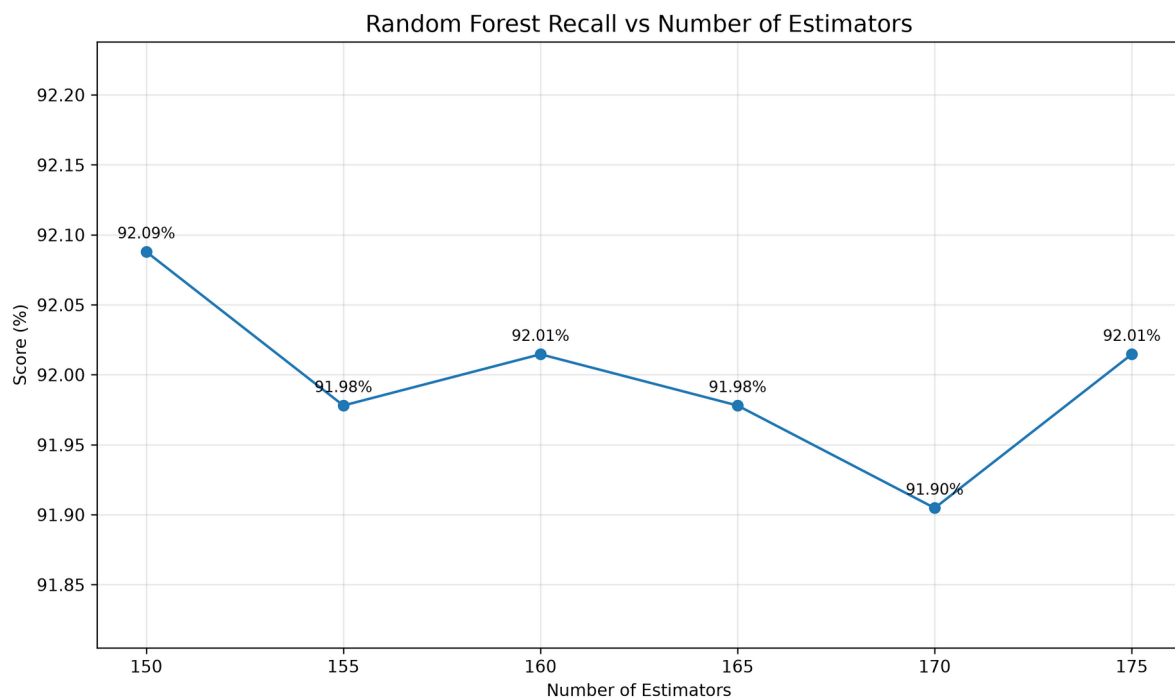
Appendix C - Estimator Tuning Tests



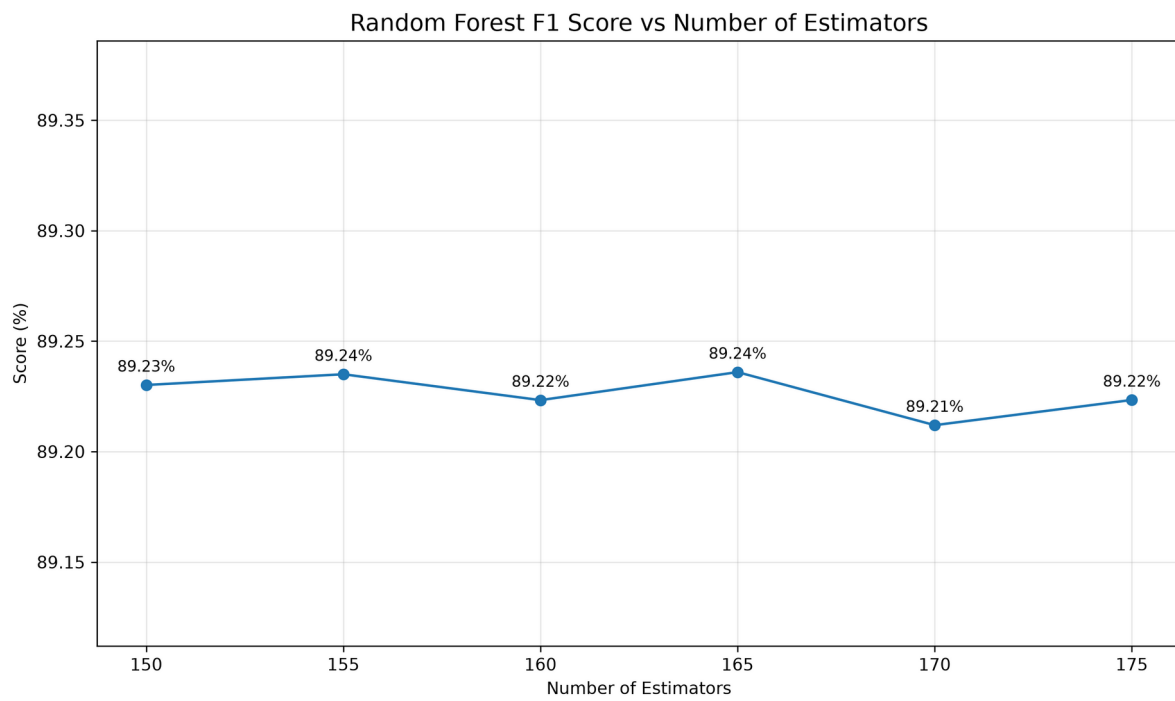
Appendix graph: Estimators Accuracy.



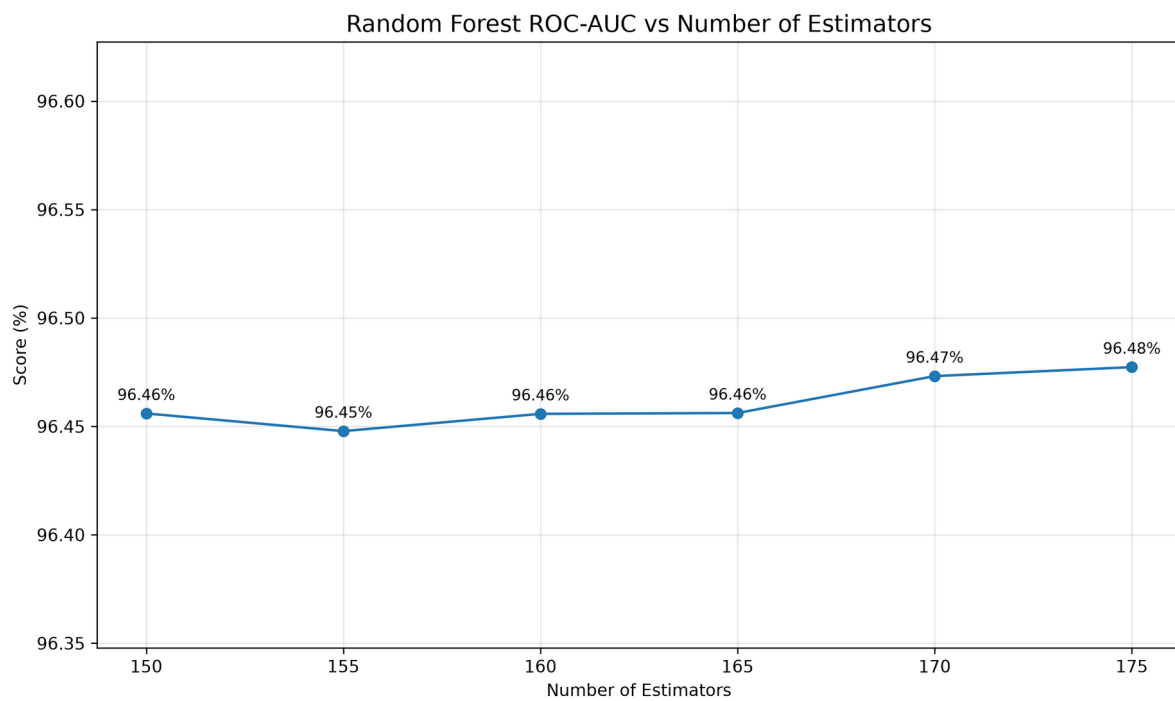
Appendix graph: Estimators Precision.



Appendix graph: Estimators Recall.

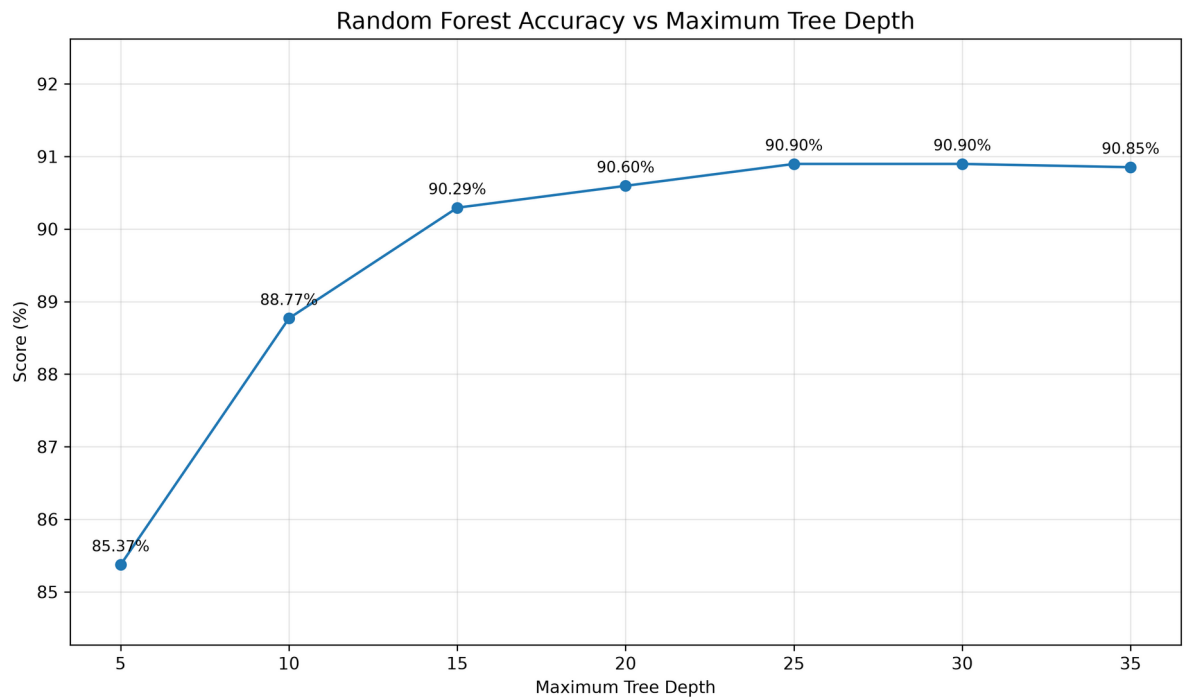


Appendix graph: Estimators F1.

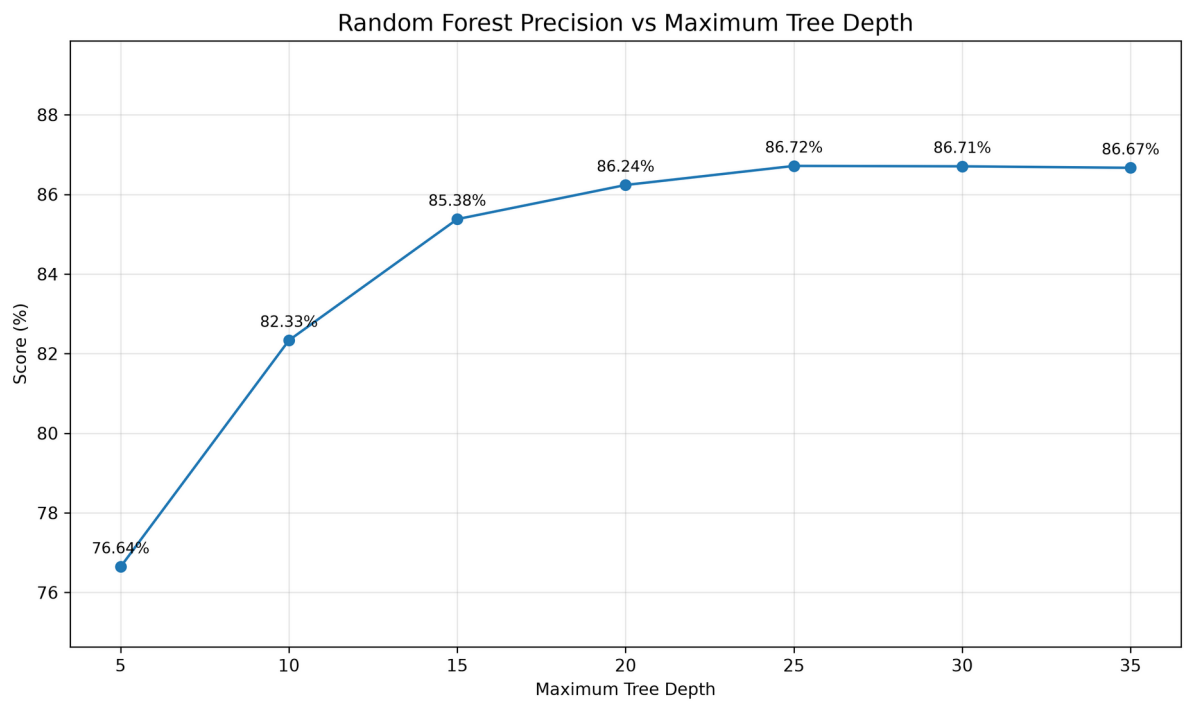


Appendix graph: Estimators Roc Auc.

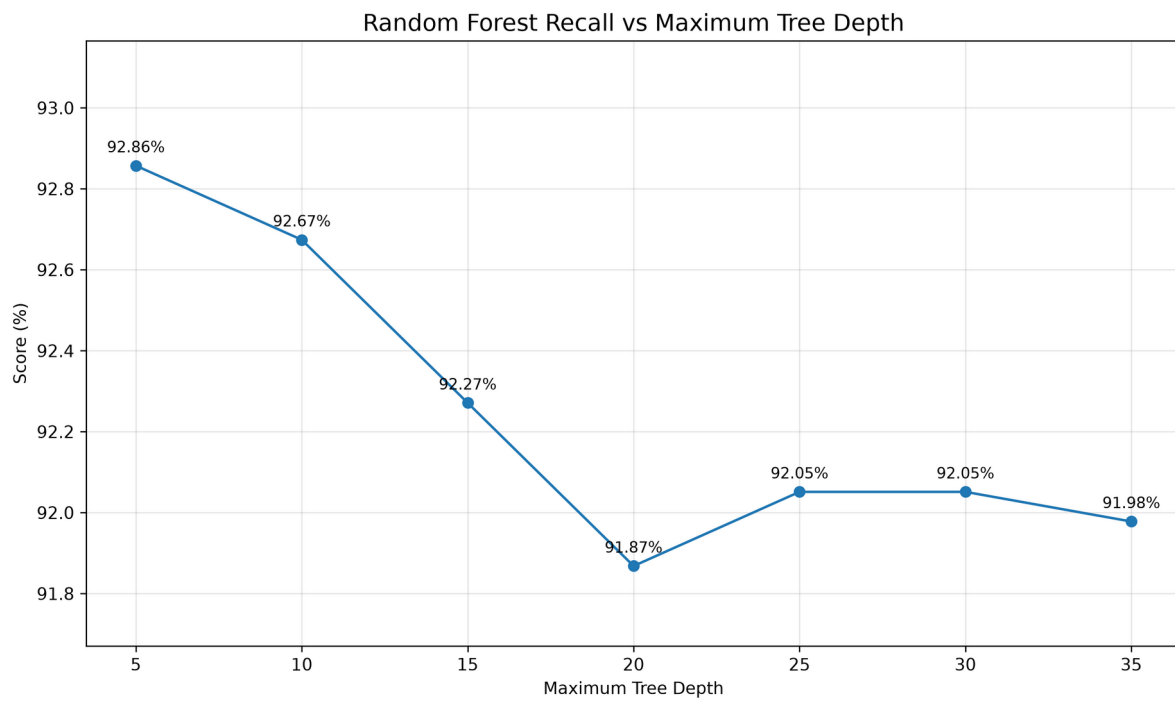
Appendix D - Depth Tests



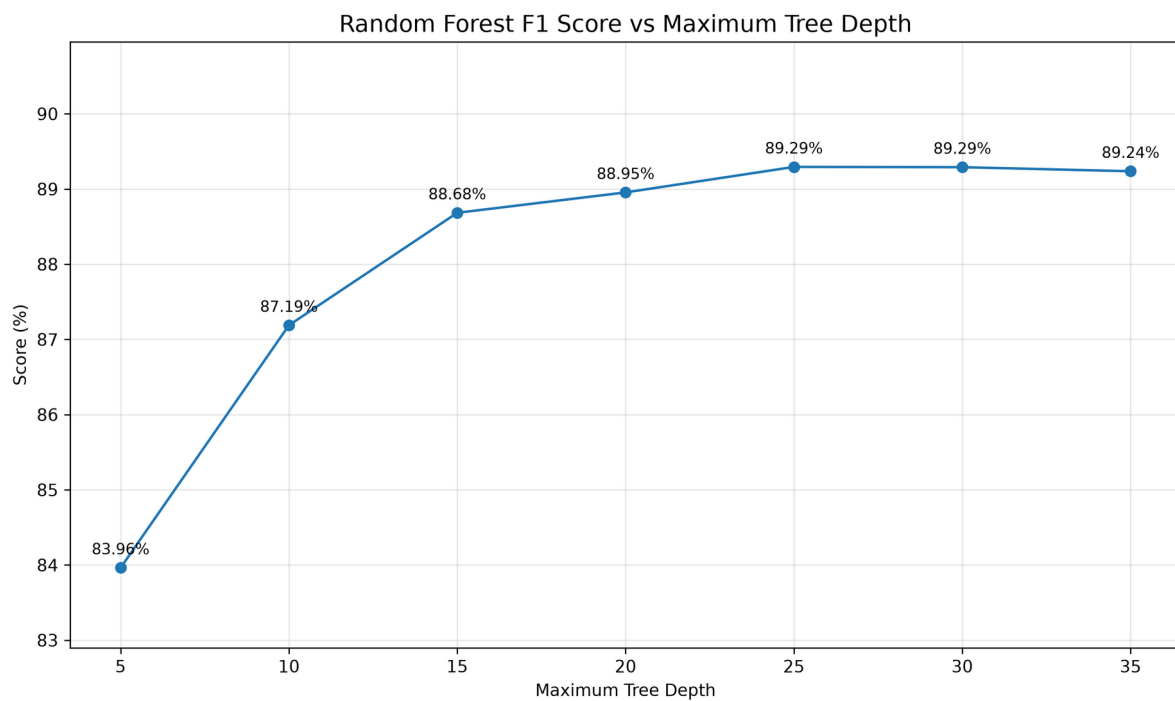
Appendix graph: Depth Accuracy.



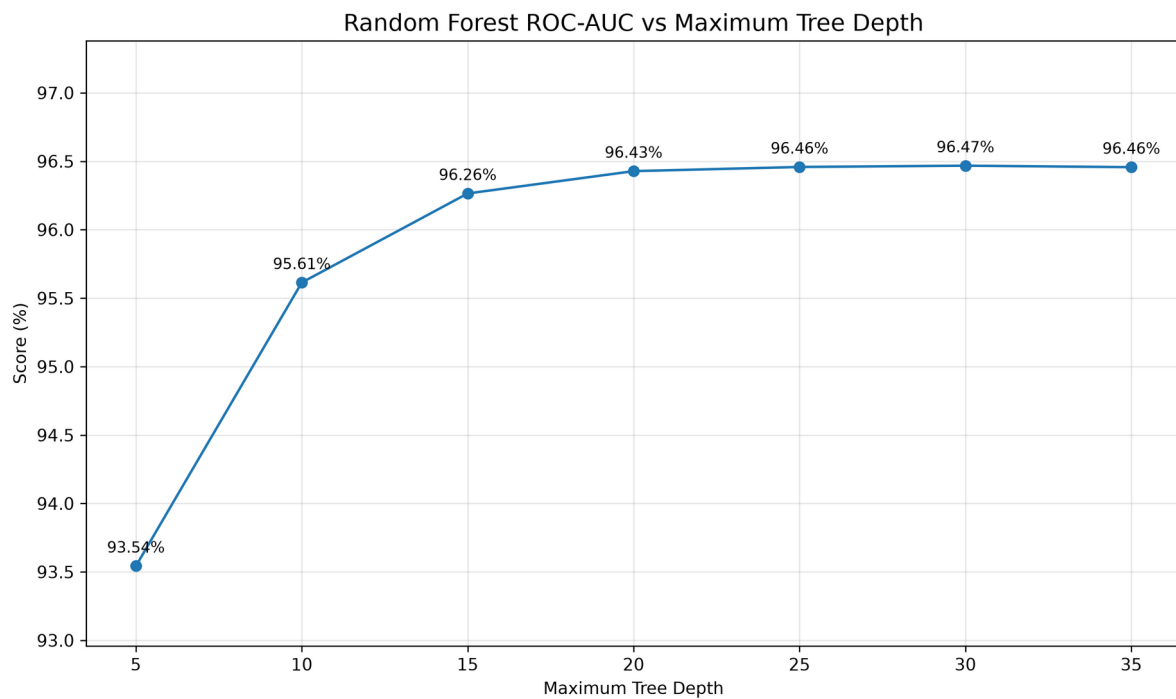
Appendix graph: Depth Precision.



Appendix graph: Depth Recall.

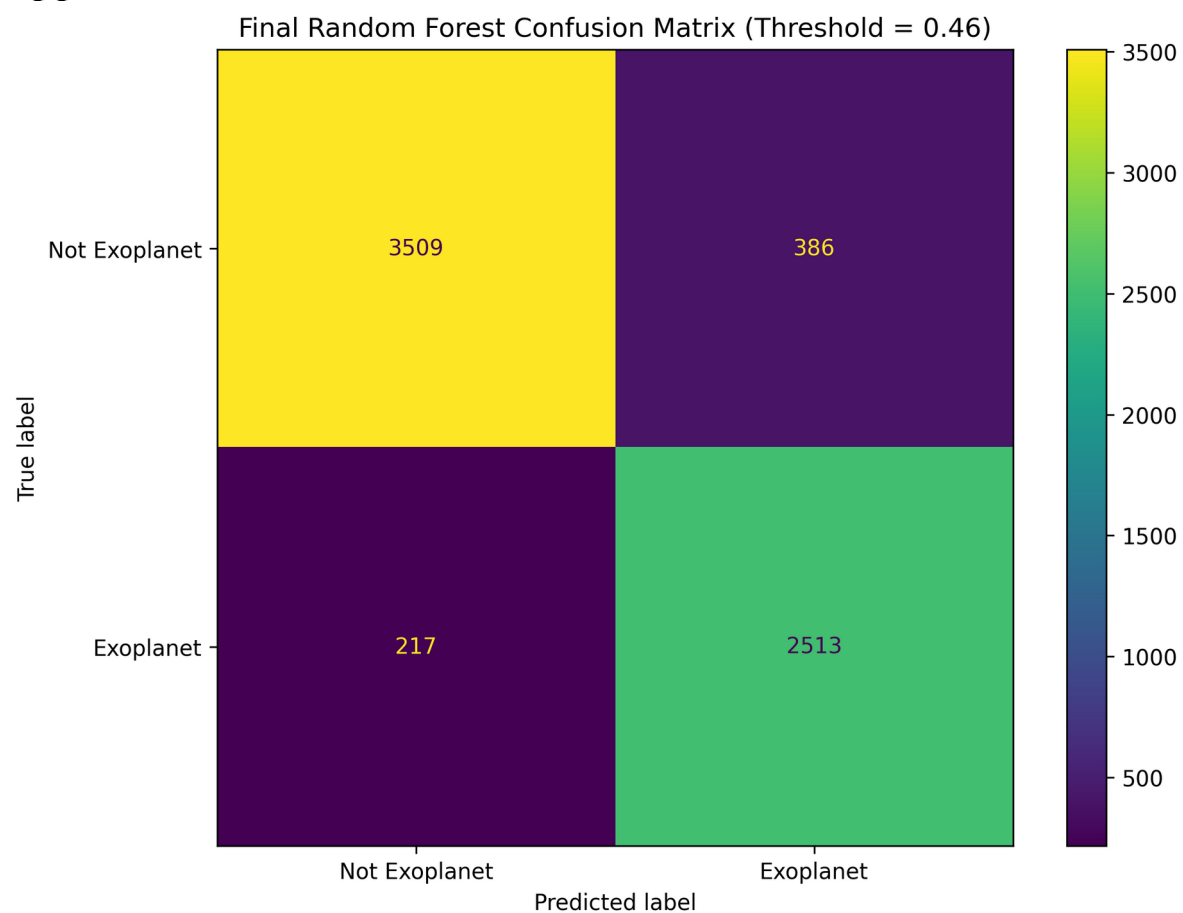


Appendix graph: Depth F1.

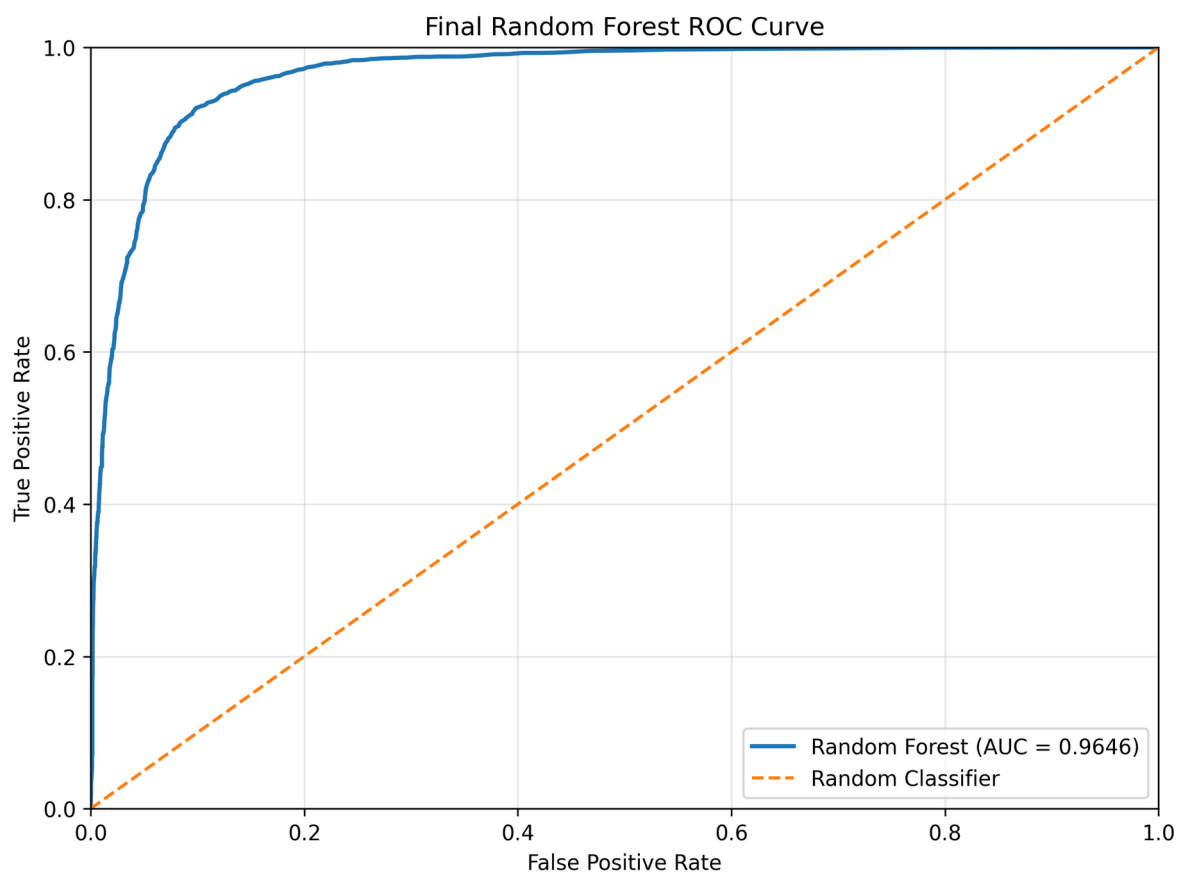


Appendix graph: Depth Roc Auc.

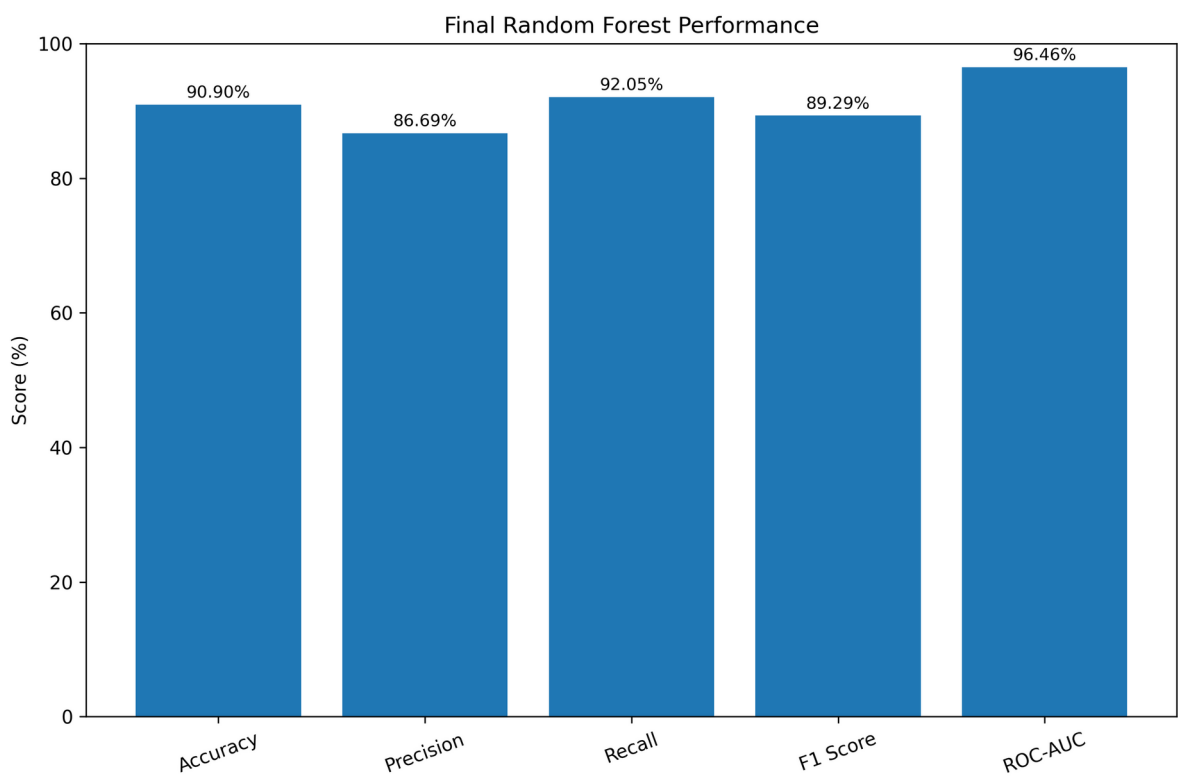
Appendix E - Final Evaluation



Appendix graph: Final Confusion Matrix.



Appendix graph: Final Roc Curve.



Appendix graph: Final Metrics.